ANALYTIC SEWING FOR CHIRAL ALGEBRAS

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Abstract. The sewing envelope $\mathcal{A}^{\mathrm{sew}}$ is the locally convex completion generated by bordism-amplitude seminorms. A positive Hilbert model with Hilbert–Schmidt sector multiplication gives trace-class closed amplitudes for q-damped sewing graphs once the caps, propagators, and loop factors satisfy the stated bounds. The Heisenberg and lattice cases provide the explicit abelian models: the Heisenberg envelope is Bergman-Fock, and the lattice envelope is the corresponding theta-weighted charge completion. For interacting families the general criterion is conditional on a positive Hilbert realisation and polynomial matrix-coefficient bounds in orthonormal bases. The coderived category D^{co} is the homological target for curved genus-g bar complexes under the stated coacyclicity hypotheses.

Contents

2020 *Mathematics Subject Classification.* 17B69 (primary); 14H10, 30H20, 47B10, 81T40 (secondary).
Key words and phrases. Chiral algebras, sewing, bar construction, Hilbert–Schmidt operators, Bergman space, Fredholm determinant, coderived category, partition functions.

1. Introduction

1.1. The algebraic bar complex and the convergence problem. The bar complex of a chiral algebra $\mathcal{A}$ is the tensor coalgebra $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with differential extracted from the operator product expansion by residues at collision diagonals in the Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n(\mathbb{C})$. The construction is purely algebraic: it uses only the OPE coefficients, the augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$, and the desuspension s^{-1} with $|s^{-1}v| = |v| - 1$. The bar differential satisfies $D^2 = 0$ on a fixed curve, and the Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0 \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ governs the modular obstruction tower.

Partition functions require more. A genus-g partition function is a trace

$$Z_g(\mathcal{A};\,\Omega) \;=\; \mathrm{Tr}_{\mathcal{A}}\big(q_1^{L_0-c/24}\cdots q_g^{L_0-c/24}\big)$$

over a state space completed in a topology for which the operator q^{L_0} is trace class. The algebraic bar complex produces formal power series in the sewing parameters; the question is whether these series converge to honest complex numbers.

This convergence problem is the analytic gap behind modular Koszul duality [5]. The five theorems (A through D and H) are algebraic statements. This paper isolates sufficient analytic hypotheses under which the formal genus expansions converge as trace-class Hilbert-space traces.

1.2. **Main results.**

1.2.0.1. The sewing envelope. For any positive-energy chiral algebra $\mathcal{A}$, we construct the *sewing envelope* $\mathcal{A}^{\mathrm{sew}}$ (Definition 2.1), the universal locally convex completion for which every bordism amplitude extends continuously, and establish its universal property (Proposition 2.2).

1.2.0.2. The Hilbert–Schmidt sewing condition. We formulate a single analytic hypothesis (Definition 3.1): the sector multiplication operators $m^c_{a,b}\colon H_a \otimes H_b \to H_c$ satisfy

$$\sum_{a,b,c\geq 0} q^{a+b+c}\,\|m^c_{a,b}\|^2_{\mathrm{HS}} \;<\; \infty \tag{1.1}$$

for some $0 < q < 1$. Under the closed-loop hypotheses of Proposition 3.2, this estimate delivers the sewing package for the q-damped kernel: every closed amplitude is trace class on the weighted completion $H_q := \widehat{\bigoplus}_n q^n H_n$.

1.2.0.3. The Heisenberg Fredholm determinant. For a rank-r Heisenberg algebra (Theorem 4.3): the sewing envelope is $\mathrm{Sym}A^2(D)$; the pair-of-pants amplitude is the second quantisation $\Gamma(R)$ of a one-particle Bergman restriction; every genus-g partition function in the standard free-boson normalisation is the Fredholm determinant

$$Z_g(\mathcal{H}^{(r)};\,\Omega) \;=\; (\det \mathrm{Im}\,\Omega)^{-r/2}\cdot(\det{}'_{\zeta}\Delta_{\Sigma_g})^{-r/2}. \tag{1.2}$$

For a single current at level k, the exponent is 1; the level rescales the two-point function and does not count oscillator species.

1.2.0.4. Lattice sewing. For a lattice vertex algebra V_Λ of rank r (Theorem 5.1): the genus-g partition function factors as

$$Z_g(V_\Lambda;\,\Omega) \;=\; Z_g(\mathcal{H}_r;\,\Omega)\cdot\Theta_\Lambda(\Omega),$$

where $\Theta_\Lambda(\Omega) = \sum_{\lambda\in\Lambda^g} e^{i\pi\lambda^T\Omega\lambda}$ is the Siegel theta function of the lattice, absolutely convergent on the Siegel upper half-space $\mathfrak{H}_g$.

1.2.0.5. The general HS-sewing theorem. Any positive-energy chiral algebra with a chosen positive Hilbert realisation, subexponential sector growth, and polynomial OPE matrix-coefficient growth satisfies the HS-sewing condition (Theorem 6.1). The criterion verifies unconditionally for the Heisenberg and lattice oscillator models treated below. For affine Kac–Moody, Virasoro, and $\mathcal{W}_N$ families it is a normed-model criterion: the required Hilbert realisation, heat trace, bounded collar transport, and polynomial matrix-coefficient estimates must be supplied.

1.3. **The three-layer gap and the coderived category.** Convergent partition functions are the beginning of the analytic theory, not its end. Three layers of structure separate the algebraic bar construction from a complete analytic modular Koszul theory:

(1) *The sewing envelope for interacting algebras.* For the Heisenberg and lattice algebras, the sewing envelope admits an explicit description (Bergman symmetric algebra, charge-refined Hilbert completion). For non-abelian chiral algebras (affine Kac–Moody at generic level, Virasoro, $\mathcal{W}_N$), the sewing envelope exists abstractly but lacks a concrete Hilbert-space identification.
(2) *Metric independence of the ind-Hilbert factorisation.* The ind-Hilbert completion of Moriwaki [6] depends a priori on a conformal metric on the curve; the algebraic bar complex does not. Identifying the two requires proving that the analytic completion is independent of the metric, a functional-analytic problem with no algebraic counterpart.
(3) *The coderived shadow at $g \geq 1$.* At genus $g \geq 1$ the bar complex carries curvature: $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g \neq 0$. The ordinary derived category is empty (every object is acyclic). The correct receptacle is the coderived category D^{co}, defined by totalising the curvature bicomplex.

In Sections 7 and 8 we develop the third layer in detail: the coderived category as the natural home for curved bar complexes, the notion of coacyclicity replacing acyclicity, and the BV/bar comparison theorem in D^{co} for all four shadow classes (G, L, C, M) of the standard landscape.

1.4. **Relation to prior work.** Segal [9] formulated the sewing axiom as the defining property of a conformal field theory. Zhu [10] proved modular invariance of genus-1 characters for rational vertex operator algebras. Huang [4] extended the sewing of sphere amplitudes to genus one and by induction to all genera, under a list of analytic convergence hypotheses. The HS criterion of this paper is strictly stronger: a single norm estimate (1.1) implies all convergences simultaneously.

Moriwaki [6] identified the Heisenberg sewing envelope with $\mathrm{Sym} A^2(D)$ through factorisation homology valued in ind-Hilbert spaces. We reinterpret this as a one-particle reduction of the pair-of-pants operator and deduce the Fredholm determinant formula (1.2).

The Belavin–Knizhnik theorem [2] expresses the bosonic string measure as a product of determinants of Laplacians; our formula (1.2) is the vertex-algebraic incarnation. Beilinson–Drinfeld [1] develop the algebraic theory of chiral algebras and factorisation; our bar complex is the explicit chain model of their factorisation coalgebra.

The coderived category for curved dg-modules was introduced by Positselski [7]; its application to curved bar complexes in the chiral setting appears in [5]. The BV/bar identification in the coderived category (Theorem 8.7) extends the genus-0 chain-level result of Costello and Gwilliam [3] to all genera.

1.5. **The modular Koszul duality context.** Analytic sewing is one component of the algebraic structure. Modular Koszul duality [5] constructs, for every chirally Koszul algebra $\mathcal{A}$, a package of algebraic invariants:

(a) the bar complex $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with differential and deconcatenation coproduct;
(b) the Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0 \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$;
(c) the shadow obstruction tower $\Theta_{\mathcal{A}} = \sum_{r\geq 2} S_r(\mathcal{A})$ with $S_2 = \kappa(\mathcal{A})$ the modular characteristic;
(d) the Koszul dual $\mathcal{A}^!$ and the complementarity relation $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = K(\mathcal{A})$;
(e) five theorems (A through D and H) controlling the structure of the bar complex, bar-cobar inversion, complementarity, the genus expansion, and chiral Hochschild cohomology.

All of this is algebraic: the bar complex is built from OPE coefficients and residues; no convergence hypothesis enters. Analytic sewing asks when the formal genus expansion $\Theta_{\mathcal{A}} = \sum_g \Theta^{(g)}_{\mathcal{A}}$ converges to a well-defined partition function $Z_g(\mathcal{A};\, \Omega)$.

The algebraic theorem gives the formal expression; the analytic criterion gives sufficient conditions for numerical convergence. The HS-sewing condition converts the formal genus expansion into a convergent trace-class stable-graph expansion. It is a Fredholm determinant only in Gaussian one-particle situations.

Remark 1.1 (The four shadow classes and sewing). The shadow classification (G, L, C, M) of the standard landscape controls the complexity of analytic sewing:

(a) *Class G* (Heisenberg, lattice): the shadow tower terminates at degree 2; sewing is fully explicit (Sections 4 and 5).
(b) *Class L* (affine Kac–Moody): the shadow tower terminates at degree 3; sewing follows from the normed HS-sewing criterion, while a concrete sewing envelope remains a separate problem.
(c) *Class C* ($\beta\gamma$, bc): the shadow tower terminates at degree 4; the BV/bar comparison is conditional on harmonic decoupling.
(d) *Class M* (Virasoro, $\mathcal{W}_N$): the shadow tower is infinite; the chain-level BV/bar comparison fails; the coderived category is essential (Section 8).

1.6. **Outline.** Section 2 constructs the sewing envelope. Section 3 formulates the Hilbert–Schmidt sewing condition and proves the trace-class theorem. Section 4 treats the decisive first case: the Heisenberg Fredholm determinant. Section 5 develops lattice sewing via theta functions. Section 6 proves the general HS-sewing theorem and verifies the criterion on the standard landscape. Section 7 analyses the three-layer analytic gap. Section 8 develops the coderived category as the correct receptacle for curved bar complexes at $g \geq 1$.

1.7. **Conventions.** All chiral algebras are $\mathbb{Z}_{\geq 0}$-graded (positive energy), with finite-dimensional weight spaces. Grading is cohomological: $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar complex uses the augmentation ideal: $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with $\bar{\mathcal{A}} = \ker(\varepsilon)$. The Dedekind eta function is $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$ with $q = e^{2\pi i\tau}$.

2. The sewing envelope

2.1. **Algebraic cores and bordism amplitudes.** An *algebraic chiral core* is a $\mathbb{Z}_{\geq 0}$-graded vector space $\mathcal{A}_{\mathrm{alg}} = \bigoplus_{n\geq 0} H_n$ equipped with the operator product expansion of a vertex operator algebra of positive energy, with $H_0 = \mathbb{C}\cdot\mathbf{1}$ and $\dim H_n < \infty$. The sector components H_n are the conformal-weight eigenspaces. The augmentation $\varepsilon\colon \mathcal{A}_{\mathrm{alg}} \to \mathbb{C}$ is the projection to H_0. The augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the input to the bar construction $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$.

A finite conformally flat bordism is a compact Riemann surface Σ with parametrised in/out boundary collars. Each such Σ determines an algebraic amplitude

$$A_\Sigma^{\mathrm{alg}}\colon \mathcal{A}_{\mathrm{alg}}^{\otimes m} \longrightarrow \mathcal{A}_{\mathrm{alg}}^{\otimes \ell},$$

a priori valued in formal series of conformal-weight components. Matrix coefficients

$$\langle \eta,\, A_\Sigma^{\mathrm{alg}}(\xi_1\otimes\cdots\otimes\xi_m)\rangle \;\in\; \mathbb{C}[[\text{moduli of }\Sigma]]$$

are the objects whose convergence analytic sewing must establish.

2.2. **Construction of the sewing envelope.**

Definition 2.1 (Sewing envelope)**.** For every finite conformally flat bordism Σ and every choice of boundary vectors $\xi_i \in \mathcal{A}_{\mathrm{alg}}$, $\eta \in \mathcal{A}_{\mathrm{alg}}^\vee$, define the seminorm

$$p_{\Sigma,\xi,\eta}(a) \;=\; \left|\langle \eta,\, A_\Sigma^{\mathrm{alg}}(\xi_1\otimes\cdots\otimes a\otimes\cdots\otimes\xi_m)\rangle\right|. \tag{2.1}$$

The *sewing envelope* $\mathcal{A}^{\mathrm{sew}}$ is the Hausdorff completion of $\mathcal{A}_{\mathrm{alg}}$ for the locally convex topology generated by all seminorms $p_{\Sigma,\xi,\eta}$.

This topology records exactly the bordism matrix coefficients used below. Continuity into this completion is therefore a separate analytic assertion, not part of the definition.

Proposition 2.2 (Universal property). *Let E be any complete locally convex realisation of $\mathcal{A}_{\mathrm{alg}}$ for which every algebraic amplitude A_Σ^{alg} extends continuously. Then the identity on $\mathcal{A}_{\mathrm{alg}}$ extends uniquely to a continuous map $E \to \mathcal{A}^{\mathrm{sew}}$ with dense image, provided $\mathcal{A}_{\mathrm{alg}}$ is dense in E.*

Proof. By hypothesis, each seminorm $p_{\Sigma,\xi,\eta}$ is continuous on E. Hence the identity $(\mathcal{A}_{\mathrm{alg}}, E\text{-topology}) \to (\mathcal{A}_{\mathrm{alg}}, \text{sewing topology})$ is continuous. Since $\mathcal{A}^{\mathrm{sew}}$ is complete, this map extends uniquely from the dense algebraic core to E. □

Remark 2.3 (Comparison with other completions). The sewing envelope need not coincide with naive completions: the q-adic completion $\widehat{\bigoplus}_n q^n H_n$, the Hilbert-space completion with respect to a fixed invariant form, Zhu's C_2-quotient, or Moriwaki's ind-Hilbert completion. They compare to $\mathcal{A}^{\mathrm{sew}}$ by continuous maps once their amplitudes are continuous; embeddings or quotient descriptions require additional topological hypotheses. The HS-sewing criterion of the next section constructs a concrete Hilbert completion H_q equipped with a dense comparison map $H_q \to \mathcal{A}^{\mathrm{sew}}$.

2.3. **Functoriality.** A grading-preserving VOA morphism $f : \mathcal{A} \to \mathcal{B}$ induces a map of sewing envelopes whenever pullback of bordism-amplitude seminorms for $\mathcal{B}$ is continuous for the sewing topology of $\mathcal{A}$. Under this seminorm-domination hypothesis the algebraic map extends uniquely by completion:

$$f^{\mathrm{sew}} : \mathcal{A}^{\mathrm{sew}} \longrightarrow \mathcal{B}^{\mathrm{sew}}.$$

Subalgebra inclusions and quotient maps therefore require a checked topological comparison; they are not automatic consequences of the algebraic inclusion.

Proposition 2.4 (Direct-sum decomposition). *Let $\mathcal{A} = \bigoplus_{\gamma\in I} \mathcal{A}_\gamma$ be a decomposition of the algebraic core into sectors preserved by the pair-of-pants multiplication. Then*

$$\mathcal{A}^{\mathrm{sew}} = \widehat{\bigoplus}_{\gamma\in I} \mathcal{A}_\gamma^{\mathrm{sew}},$$

where the completion is in the product topology.

Proof. Each seminorm $p_{\Sigma,\xi,\eta}$ decomposes by sector selection (insertion of projection operators onto the sector γ at each boundary circle); the cross-terms are separate seminorms. The completion of a direct sum in a product of locally convex topologies is the product of completions. □

2.4. **Comparison of analytic hypotheses.** The algebraic bar differential, positive-energy finiteness, Koszulness, HS-sewing, and modular-functor finiteness are logically distinct hypotheses. This paper uses only the implications stated at the point where they enter the proofs. In particular, HS-sewing does not imply Koszulness, and C_2-cofiniteness does not by itself give a Hilbert–Schmidt pair-of-pants estimate.

Remark 2.5 (Standard families). The Heisenberg and lattice oscillator models provide explicit positive Hilbert realisations. For affine Kac–Moody, Virasoro, and $\mathcal{W}_N$ families, the HS-sewing theorem below is a normed criterion: one must fix a positive Hilbert model and verify the heat trace, bounded collar transport, and polynomial matrix-coefficient bounds in that model.

2.5. **The bar complex as a family over moduli.** The bar complex at genus g is a family of cochain complexes parametrised by $\overline{\mathcal{M}}_g$. On the smooth locus $\mathcal{M}_g$ the differential satisfies $d^2 = 0$; on the compactification $\overline{\mathcal{M}}_g$ the differential acquires curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ from the boundary strata.

The Gauss–Manin connection $\nabla^{\mathrm{GM}} : B^{(g)}(\mathcal{A}) \to B^{(g)}(\mathcal{A}) \otimes \Omega^1_{\overline{\mathcal{M}}_g}$ controls the variation of the bar complex with moduli. The total differential $D_g = d_{\mathrm{fib}} + \nabla^{\mathrm{GM}}$ satisfies $D_g^2 = 0$: the fiberwise curvature $d_{\mathrm{fib}}^2 = \kappa\omega_g$ is exactly cancelled by the Gauss–Manin curvature $(\nabla^{\mathrm{GM}})^2 = -\kappa\omega_g$.

Analytic sewing completes each fixed fiber to a Hilbert space. A continuous Gauss–Manin extension is a separate analytic estimate on the variation of prime forms, collars, and Quillen curvature.

Proposition 2.6 (Conditional Gauss–Manin continuity). *Assume HS-sewing and, in addition, uniform H_q-operator bounds for the moduli derivatives of the sewing kernels on the chosen compact collar chart. Then ∇_g^{GM} extends to a bounded operator on the weighted completion H_q, and the total differential $D_g = d_{\mathrm{fib}} + \nabla_g^{\mathrm{GM}}$ satisfies $D_g^2 = 0$ on the completed bar complex.*

Proof. The Gauss–Manin connection acts on bar elements by variation of the bar propagator $d\log E(z,w)$ with respect to moduli. By the additional hypothesis these variations define bounded operators on H_q. The identity $D_g^2 = 0$ is algebraic on the dense core and extends by continuity. □

3. The Hilbert–Schmidt sewing condition

The universal sewing envelope of Definition 2.1 is an abstract object. Computation requires a concrete Hilbert completion with bounded sewing operators whose loop compositions are trace class.

3.1. The condition.

Definition 3.1 (HS-sewing). Let $\mathcal{A}_{\mathrm{alg}} = \bigoplus_n H_n$ be an algebraic chiral core with a pre-Hilbert structure on each sector (for example, the restriction of the invariant bilinear form to H_n). Write the pair-of-pants sector multiplications as

$$m^c_{a,b} \colon H_a \otimes H_b \longrightarrow H_c.$$

Say $\mathcal{A}$ satisfies *HS-sewing* at parameter $q \in (0,1)$ if

$$\sum_{a,b,c\geq 0} q^{a+b+c} \, \|m^c_{a,b}\|^2_{\mathrm{HS}} \; < \; \infty. \tag{3.1}$$

The *weighted completion* is $H_q := \widehat{\bigoplus}_{n\geq 0} q^n H_n$, with inner product $\langle v,w\rangle_q = q^{2n}\langle v,w\rangle_{H_n}$ for $v,w \in H_n$. The *q-damped pair-of-pants* is the sectorwise map

$$m_q\big|_{H_a\otimes H_b\to H_c} \; := \; q^{(3a+3b-c)/2}\, m^c_{a,b},$$

viewed as an operator $H_q \widehat{\otimes} H_q \to H_q$.

The damping is part of the sewing kernel. The raw multiplication on the Hilbert space with inner product $q^{2n}\langle -,-\rangle_{H_n}$ contributes the sector factor $q^{2(c-a-b)}$ to the Hilbert–Schmidt norm, not the factor in (3.1).

3.2. The trace-class theorem.

Proposition 3.2 (Closed q-damped loop amplitudes are trace class). *Suppose $\mathcal{A}$ satisfies HS-sewing at parameter q. Then:*

(i) *the q-damped pair-of-pants $m_q \colon H_q \otimes H_q \to H_q$ is Hilbert–Schmidt, with*

$$\|m_q\|^2_{\mathrm{HS}} \; = \; \sum_{a,b,c} q^{a+b+c}\, \|m^c_{a,b}\|^2_{\mathrm{HS}};$$

(ii) *under bounded collar transport and the heat-trace condition $\sum_n q^n \dim H_n < \infty$, every closed stable amplitude built from a finite pants decomposition with at least two Hilbert–Schmidt pair-of-pants factors is trace class on H_q;*

(iii) *if every open sewing seminorm $p_{\Sigma,\xi,\eta}$ is separately continuous on H_q, then the weighted Hilbert completion H_q admits a continuous linear map $H_q \to \mathcal{A}^{\mathrm{sew}}$ whose image is dense.*

Proof. Claim (i): identifying $H_a \otimes H_b$ with the weight-$(a+b)$ subspace of $H_q^{\widehat{\otimes}2}$, the raw sector component $m^c_{a,b}$ contributes $q^{2(c-a-b)}\|m^c_{a,b}\|^2_{\mathrm{HS}}$. The damping $q^{(3a+3b-c)/2}$ multiplies this by $q^{3a+3b-c}$, so the sector contribution is $q^{a+b+c}\|m^c_{a,b}\|^2_{\mathrm{HS}}$. Summing gives $\|m_q\|^2_{\mathrm{HS}} = \sum q^{a+b+c}\|m^c_{a,b}\|^2_{\mathrm{HS}} < \infty$.

Claim (ii) is the standard fact that a composition of two Hilbert–Schmidt operators is trace class, together with the stated bounded collar transport and heat-trace assumptions. Thus a closed stable sewing graph with the controlled factors has a well-defined trace. Open amplitudes and loops with only one controlled factor require separate uniform boundedness assumptions.

Claim (iii) is a conditional comparison statement. If the open sewing seminorms of Definition 2.1 are continuous on H_q, then the identity $(\mathcal{A}_{\mathrm{alg}}, H_q\text{-topology}) \to (\mathcal{A}_{\mathrm{alg}}, \text{sewing topology})$ follows, and completion gives $H_q \to \mathcal{A}^{\mathrm{sew}}$. □

Corollary 3.3 (Partition functions converge under closed graph bounds). *Under the hypotheses of Proposition 3.2, the genus-g partition function $Z_g(\mathcal{A};\,\Omega) = \mathrm{Tr}(q_1^{L_0 - c/24} \cdots q_g^{L_0 - c/24})$ converges absolutely for all period matrices $\Omega \in \mathfrak{H}_g$ with $|q_j| = |e^{2\pi i \Omega_{jj}}| < q$, producing a holomorphic function on a domain in the Siegel upper half-space.*

Proof. The partition function is a closed amplitude (trace over the state space with g sewing circles). By Proposition 3.2(ii), it is the trace of a trace-class operator on H_q, hence absolutely convergent. Holomorphicity in the moduli follows from the absolute convergence and the holomorphic dependence of the sewing kernel on Ω. □

Corollary 3.4 (Gram positivity under reflection positivity). *Under HS-sewing together with a unitary reflection-positive Hilbert model and positive sewing operators, the connected sewing amplitudes $a_{\mathcal{A}}(N) \in \mathbb{R}_{\geq 0}$ define a positive-semidefinite inner product on generating Dirichlet polynomials.*

Proof. Trace-class operators form a two-sided ideal with a well-defined trace, positive when applied to a positive operator; the sewing amplitude is a composition of positive pair-of-pants operators, and positivity passes through. □

3.3. **The sewing Gram matrix and its determinant.** The Hilbert–Schmidt bound controls convergence and spectral asymptotics. Let K_g denote the sewing operator (composition of pair-of-pants restrictions around a cycle of the genus-g surface). The connected free energy is

$$F_g^{\mathrm{conn}}(\mathcal{A};\,q) \;=\; -\log\det(1 - K_g),$$

convergent whenever $\|K_g\|_1 < 1$. The eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$ of K_g satisfy $\sum_j \lambda_j < \infty$ (trace class), and the Fredholm determinant $\det(1 - K_g) = \prod_j (1 - \lambda_j)$ converges absolutely. For the Heisenberg, the eigenvalues are q^n ($n \geq 1$) and the Fredholm determinant is the Dedekind eta function (see Section 4).

3.4. **The HS-sewing condition and the shadow classes.** The HS-sewing estimate requires a normed polynomial bound on OPE matrix coefficients. The shadow class records the formal pole profile, but it does not by itself prove the Hilbert-space bound.

Proposition 3.5 (Shadow class and OPE growth input). *For each standard family, the following formal pole profile gives the matrix-coefficient estimate that must be verified in the chosen Hilbert norm:*

Class	*Families*	N	*Mechanism*
G	*Heisenberg, free fields*	1	*quadratic OPE*
L	*affine Kac–Moody*	1	*linear current algebra*
C	$\beta\gamma$, bc	1	*first-order systems*
M	*Virasoro,* $\mathcal{W}_N$	*family-dependent*	*higher-spin fields*

Proof. For class G, the OPE $J(z)J(w) \sim k/(z-w)^2$ has no field on the right-hand side beyond the central term; the structure constants are bounded by the level k, independent of weight.

For class L, the OPE $J^a(z)J^b(w) \sim k\delta^{ab}/(z-w)^2 + f^{ab}{}_c J^c(w)/(z-w)$ has at most one derivative insertion. In a normed model this gives the required bound only after the current modes are controlled in orthonormal bases.

For class C, the $\beta\gamma$ system has OPE $\beta(z)\gamma(w) \sim 1/(z-w)$, a simple pole; all higher correlators factor through this pairing by Wick's theorem.

For class M, the Virasoro OPE is quartic, and normal-ordered products of T with itself produce structure constants growing as $(n+1)^2$ in the PBW monomial norm. For $\mathcal{W}_N$, the spin-s generators require the corresponding PBW estimate; positivity and Hilbert boundedness are additional analytic inputs. □

Once the normed bound $|C^{c,k}_{a,i;\,b,j}| \leq K(a+b+c+1)^N$ is known, the HS-sewing sum is bounded by

$$\sum_{a,b,c} q^{a+b+c} \|m^c_{a,b}\|^2_{\mathrm{HS}} \;\leq\; 3^{2N} K^2 \Big(\sum_n q^n \dim H_n (n+1)^{2N}\Big)\Big(\sum_n q^n \dim H_n\Big)^2,$$

so the exponential decay of q^n dominates the polynomial factor under subexponential sector growth.

3.5. **Quantitative HS bounds.** The HS-sewing condition provides not just qualitative convergence but quantitative bounds on partition functions.

Proposition 3.6 (Partition function bound)**.** *Let $\mathcal{A}$ satisfy HS-sewing at parameter q, and fix a closed stable sewing graph whose collar transports are bounded and whose loops contain the controlled Hilbert–Schmidt factors required in Proposition 3.2. Then the corresponding closed amplitude satisfies*

$$|A_\Gamma(\mathcal{A};\,\Omega)| \;\leq\; C_\Gamma \left(\|m_q\|_{\mathrm{HS}}\right)^{N_\Gamma},$$

where C_Γ is the product of the bounded collar norms and N_Γ is the number of controlled pair-of-pants factors in the chosen decomposition.

Proof. Apply the ideal inequalities $\|AB\|_1 \leq \|A\|_{\mathrm{HS}}\|B\|_{\mathrm{HS}}$ and $\|TAB\|_1 \leq \|T\|_{\mathrm{op}}\|A\|_{\mathrm{HS}}\|B\|_{\mathrm{HS}}$ along the finite graph. The bounded collar transports contribute C_Γ and the controlled pair-of-pants factors contribute the displayed power of $\|m_q\|_{\mathrm{HS}}$. □

Corollary 3.7 (Growth in genus)**.** *For any family of closed stable graphs with uniformly bounded collar norms and linearly many controlled sewing factors in genus, the closed amplitudes grow at most exponentially in genus. The base of the exponential is determined by the chosen Hilbert model and sewing parameters.*

3.6. **Optimality of the HS condition.** The HS-sewing condition is a sufficient trace-class criterion. Two weaker conditions also imply convergence in special cases:

(a) *C_2-cofiniteness:* Zhu's condition [10] gives modular control of genus-1 characters in the rational setting. It does not by itself prove a Hilbert–Schmidt estimate for pair-of-pants multiplication.
(b) *Closed-amplitude trace class:* one could require trace-class convergence only for the closed amplitudes entering a given partition function. This can be weaker than controlling the full pair-of-pants operator, but it is less local and is harder to verify from OPE data alone.

The HS condition is local enough to be checked from OPE data once a positive Hilbert model and normed matrix-coefficient bounds are fixed.

4. The Heisenberg sewing: Fredholm determinant

The Heisenberg vertex algebra $\mathcal{H}_k$ is the basic explicit case. Every sewing operation reduces to Fock-space combinatorics and Bergman-space functional analysis.

4.1. **The Heisenberg algebra.** The Heisenberg vertex algebra $\mathcal{H}_k$ at level k is generated by a single current $J(z)$ with operator product expansion

$$J(z)\,J(w) \;\sim\; \frac{k}{(z-w)^2}.$$

The sectors H_n are spanned by monomials $a_{-n_1-1}\cdots a_{-n_j-1}\mathbf{1}$ with $\sum(n_i+1)=n$; the sector generating function is $\sum_{n\geq 0}\dim H_n\, q^n = \prod_{n\geq 1}(1-q^n)^{-1}$, equivalently $q^{1/24}/\eta(q)$. The Heisenberg is the simplest member of the standard landscape, class G in the shadow classification (the shadow tower terminates at degree 2), with $\kappa(\mathcal{H}_k)=k$.

4.2. **The Heisenberg OPE and HS-sewing verification.** The HS-sewing condition for the Heisenberg is verified directly from the OPE. The pair-of-pants multiplication $m^c_{a,b}\colon H_a\otimes H_b\to H_c$ is nonzero only when $c\leq a+b$ (the weight cannot increase beyond the total input weight). The OPE $J(z)J(w)\sim k/(z-w)^2$ produces structure constants bounded by k: the n-th mode a_n satisfies $[a_m,a_n]=km\delta_{m+n,0}$, so the only nontrivial sector multiplication is the Wick contraction $m^{a+b-1}_{a,b}\colon a_{-a-1}\mathbf{1}\otimes a_{-b-1}\mathbf{1}\mapsto k(a+1)\delta_{b,0}\,a_{-a-1}\mathbf{1}$ (plus normal-ordered products). The Hilbert–Schmidt norm is

$$\|m^c_{a,b}\|^2_{\mathrm{HS}} \;\leq\; k^2 p(a)p(b)p(c)(a+b+c+1)^2,$$

and the HS-sewing sum is

$$\sum_{a,b,c} q^{a+b+c}\|m^c_{a,b}\|^2_{\mathrm{HS}} \;\leq\; 9k^2\Big(\sum_n q^n p(n)(n+1)^2\Big)\Big(\sum_n q^n p(n)\Big)^2.$$

Each series converges for $0<q<1$ because $p(n)$ grows subexponentially ($\log p(n)\sim\pi\sqrt{2n/3}$) while q^n decays exponentially. The radius of convergence $q=1$ is sharp: at $q=1$ the series $\sum p(n)=\infty$.

Remark 4.1 (The partition number and subexponential growth). The Hardy–Ramanujan asymptotic $p(n)\sim\frac{1}{4n\sqrt{3}}\exp\big(\pi\sqrt{2n/3}\big)$ shows that $\log p(n)=O(\sqrt{n})$, hence $p(n)\leq C\exp(\alpha\sqrt{n})$ for appropriate C and α. The exponential decay $q^n=\exp(-n\log(1/q))$ dominates the subexponential growth for any $q<1$: $q^n p(n)\leq C\exp(\alpha\sqrt{n}-n\log(1/q))$, which decays superpolynomially. This is the mechanism behind the HS-sewing convergence for all standard families.

4.3. **The Bergman space.** Let $A^2(D)$ be the Bergman space of square-integrable holomorphic functions on the unit disk $D\subset\mathbb{C}$, with inner product $\langle f,g\rangle=\int_D f\overline{g}\,dA$ and orthonormal basis $e_n(z)=\sqrt{(n+1)/\pi}\,z^n$ for $n\geq 0$. The symmetric algebra $\mathrm{Sym}A^2(D)$ is the bosonic Fock space built on the one-particle Bergman space. The Bergman reproducing kernel is

$$K_D(z,w) \;=\; \frac{1}{\pi(1-z\bar{w})^2}.$$

Definition 4.2 (Mode–Bergman map). The *mode–Bergman map* $\Theta\colon\mathcal{H}^{\mathrm{alg}}_k\to\mathrm{Sym}A^2(D)$ is the dense linear map defined on the algebraic Fock core by

$$\begin{aligned}\Theta(a_{-n-1}\mathbf{1}) \;&=\; \sqrt{n+1}\,z^n,\\ \Theta(a_{-n_1-1}\cdots a_{-n_j-1}\mathbf{1}) \;&=\; \mathrm{Sym}\big(\Theta(a_{-n_1-1}\mathbf{1})\otimes\cdots\otimes\Theta(a_{-n_j-1}\mathbf{1})\big).\end{aligned}$$

The map Θ intertwines the algebraic Fock space grading with the polynomial-degree grading of the Bergman space and sends the invariant bilinear form on $\mathcal{H}_k$ to the Bergman inner product (up to the level factor k).

4.4. The Heisenberg sewing theorem.

Theorem 4.3 (Heisenberg sewing)**.** *Let $\mathcal{H}^{(r)}$ be the rank-r Heisenberg vertex algebra and let Θ be the mode–Bergman map.*

(i) *The sewing envelope $(\mathcal{H}^{(r)})^{\mathrm{sew}}$ is identified via Θ with* $\mathrm{Sym}\, A^2(D)$.
(ii) *The completed bar differential on Bergman tensors is the closure of the Gaussian collision operator.*
(iii) *The pair-of-pants sewing amplitude is the second quantisation $\Gamma(R)$ of a one-particle Bergman restriction $R\colon A^2(D_{\mathrm{out}}) \to A^2(D_{\mathrm{in},1}) \otimes A^2(D_{\mathrm{in},2})$.*
(iv) *Every closed genus-g amplitude is a Fredholm determinant on the reduced one-particle Bergman space $A^2_0(\partial D)$ (constant mode removed).*

Proof. Claim (i) is the ind-Hilbert identification of Moriwaki [6], who constructs a conformally flat 2-disk algebra structure on $\mathrm{Sym}\, A^2(D)$ isomorphic to the Heisenberg sewing completion by factorisation homology valued in IndHilb. The universal property (Proposition 2.2) identifies his completion with the Heisenberg sewing envelope.

Claim (ii) is the analytic form of Wick contraction. At the one-particle level the current two-point function becomes the Bergman kernel under the mode–Bergman identification. The level rescales this pairing; the rank counts independent oscillator copies.

Claims (iii) and (iv) follow from the one-particle sewing theorem below. □

4.5. One-particle sewing and the Fredholm determinant.

The Heisenberg amplitude factorises completely through a one-particle operator. This is the analytic avatar of Wick's theorem.

Theorem 4.4 (Conditional one-particle sewing)**.** *The Heisenberg pair-of-pants amplitude is the second quantisation $\Gamma(R)$ of the one-particle Bergman restriction $R\colon A^2(D_{\mathrm{out}}) \longrightarrow A^2(D_{\mathrm{in},1}) \otimes A^2(D_{\mathrm{in},2})$, with matrix elements satisfying*

$$|R_{n,m}| \;\le\; C q^{(n+m)/2},$$

where $q = e^{-t}$ and t is the conformal length of the collar. The sewing operator T_q on the reduced Bergman space $A^2_0(\partial D)$ has eigenvalues q^n for $n \ge 1$ and is trace class with

$$\|T_q\|_1 \;=\; \frac{q}{1-q}.$$

When the bosonic second quantization is trace class, the genus-g partition function of a rank-r Heisenberg VOA is the Fredholm determinant

$$Z_g(\mathcal{H}^{(r)};\, \Omega) \;=\; (\det \operatorname{Im}\Omega)^{-r/2} \cdot (\det{}'_{\zeta} \Delta_{\Sigma_g})^{-r/2}.$$

At genus 1, $Z_1 = (\operatorname{Im}\tau)^{-r/2}\, |\eta(\tau)|^{-2r}$.

Proof. Wick's theorem: every Heisenberg amplitude factors into pairwise contractions of the generating current. Each contraction across a sewing circle corresponds to a one-particle pairing, so the full Fock-space amplitude equals the second quantisation $\Gamma(R)$ of the one-particle pairing R.

The one-particle matrix element is

$$R_{n,m} \;=\; \frac{k}{2\pi i} \oint_{|z|=1} \oint_{|w|=1} z^{-n-1}\, w^{-m-1}\, G_{\mathrm{pants}}(z,w)\, dz\, dw,$$

where G_{pants} is the Bergman kernel of the pair of pants. The collar of length $t = -\log q$ and the Bergman reproducing property give $|G_{\mathrm{pants}}(z,w)| \le C(1 - q|zw|)^{-2}$; extracting the (n,m) Fourier coefficient yields $|R_{n,m}| \le C q^{(n+m)/2}$.

For a trace-class one-particle operator T with spectral radius less than 1, bosonic second quantization satisfies $\mathrm{Tr}_{\mathrm{Sym}H}\,\Gamma(T) = \det(1-T)^{-1}$. For rank r, the determinant is raised to the power $-r$. A Hilbert–Schmidt one-particle restriction alone is not enough for this Fredholm formula; the theorem assumes trace-class second-quantization.

The Polyakov–Belavin–Knizhnik formula [2] identifies the Heisenberg partition function: $\det(1-K_g)^{-r} = (\det \operatorname{Im}\Omega)^{-r/2}(\det'_\zeta \Delta_{\Sigma_g})^{-r/2}$, where K_g is the genus-g sewing kernel assembled by composing $2g-2$ pair-of-pants restriction operators along the edges of a trivalent graph Γ of first Betti number g.

At genus 1, $K_1 = T_q$ and the Fredholm determinant is $\det(1-T_q) = \prod_{n\geq 1}(1-q^n) = q^{-1/24}\eta(q)$, so $Z_1 = (\operatorname{Im}\tau)^{-r/2}\cdot|\eta(\tau)|^{-2r}$ for rank r after combining holomorphic and antiholomorphic sectors. For a single current at level k, this exponent is 1; the level affects the two-point normalization, not the number of oscillator species. □

Remark 4.5 (Genus-1 connected free energy). At genus 1, the connected free energy is

$$F_1^{\mathrm{conn}}(\mathcal{H}^{(1)};\,q) \;=\; \log\det(1-T_q)^{-1} \;=\; \log\big(q^{-1/24}\eta(q)\big)^{-1} \;=\; -\frac{1}{24}\log q - \sum_{n\geq 1}\log(1-q^n),$$

the Fourier expansion of which produces the oscillator determinant coefficients $a_{\mathcal{H}^{(1)}}(N) = \sum_{d|N} 1/d$ for a single current. In the chiral shadow normalization, insertion of the current two-point level multiplies these scalar amplitudes by k. The rank-one determinant Dirichlet series is $S_{\mathcal{H}^{(1)}}(u) = \zeta(u)\,\zeta(u+1)$. The Gram positivity of Corollary 3.4 is a consequence of $a_{\mathcal{H}_k}(N) > 0$.

4.6. **Higher-genus Fredholm expansion.** The genus-g partition function is assembled by choosing a pants decomposition of Σ_g: a trivalent graph Γ with g loops (first Betti number), $2g-2$ vertices (pairs of pants), and $3g-3$ internal edges (sewing circles). Each vertex contributes a pair-of-pants restriction operator; each edge contributes a sewing operator T_q (the restriction to a collar of conformal length $t = -\log q$).

For a fixed decomposition, the genus-g sewing kernel is the composition

$$K_g \;=\; \prod_{\text{edges of }\Gamma} T_{q_e} \;\circ\; \prod_{\text{vertices of }\Gamma} R_v,$$

where $q_e = e^{-t_e}$ is the sewing parameter of the e-th internal edge.

Proposition 4.6 (Trace-class sewing kernel). *Assume the one-particle sewing kernel K_g is trace class and has spectral radius less than* 1*. Then:*

(i) *the Fredholm determinant* $\det(1-K_g)$ *is defined;*
(ii) *the power traces* $\mathrm{Tr}(K_g^m)$ *converge absolutely;*
(iii) *no eigenvalue asymptotic is used in the arguments below.*

Proof. These are standard properties of trace-class operators and Fredholm determinants. The trace-class hypothesis is supplied in the Heisenberg setting by the one-particle determinant theorem, not by a general Schottky eigenvalue estimate. □

Corollary 4.7 (Fredholm expansion at genus g). *The genus-g partition function admits the absolutely convergent Fredholm expansion*

$$Z_g(\mathcal{H}^{(r)};\,\Omega) \;=\; \det(1-K_g)^{-r} \;=\; \exp\Big(r\sum_{m\geq 1}\frac{1}{m}\mathrm{Tr}(K_g^m)\Big),$$

where the power traces $\mathrm{Tr}(K_g^m)$ *are convergent sums over products of sewing eigenvalues.*

Proof. The Fredholm determinant of a trace-class operator satisfies

$$\log\det(1-K) = -\sum_{m\geq 1}\frac{1}{m}\mathrm{Tr}(K^m),$$

convergent whenever $\|K\|_1 < 1$. The trace-class property of K_g (Proposition 4.6) ensures convergence; raising to the r-th power accounts for the number of independent oscillator species. □

4.7. **Genus-2 specialisation.** At genus 2, every Riemann surface admits a hyperelliptic representation $y^2 = \prod_{i=1}^{6}(x-e_i)$, and the period matrix is a 2×2 symmetric matrix $\Omega \in \mathfrak{H}_2$ with $\mathrm{Im}\,\Omega > 0$. The Heisenberg partition function becomes

$$Z_2(\mathcal{H}^{(r)};\,\Omega) \;=\; (\det\mathrm{Im}\,\Omega)^{-r/2}\cdot(\det{}'_{\zeta}\Delta_{\Sigma_2})^{-r/2}.$$

The 2×2 determinant $\det\mathrm{Im}\,\Omega = (\mathrm{Im}\,\Omega_{11})(\mathrm{Im}\,\Omega_{22}) - (\mathrm{Im}\,\Omega_{12})^2$ is the volume of the Jacobian torus $\mathbb{C}^2/(\mathbb{Z}^2+\Omega\mathbb{Z}^2)$.

The sewing kernel at genus 2 is assembled from a trivalent graph with 2 vertices and 3 internal edges: $K_2 = T_{q_1}\circ R_1\circ T_{q_2}\circ R_2\circ T_{q_3}$, where q_1, q_2, q_3 are the three sewing parameters corresponding to the three handles of the genus-2 surface (two non-separating, one separating). The degeneration limits $q_i \to 0$ recover the genus-1 partition functions of the constituent tori.

Remark 4.8 (Factorisation at the separating node). In the separating degeneration (one of the three sewing parameters approaches zero while the other two remain generic), the genus-2 partition function factors:

$$Z_2(\mathcal{H}^{(r)};\,\Omega) \;\xrightarrow{q_3\to 0}\; Z_1(\mathcal{H}^{(r)};\,\tau_1)\cdot Z_1(\mathcal{H}^{(r)};\,\tau_2)\cdot(1+O(q_3)),$$

where τ_1, τ_2 are the modular parameters of the two genus-1 components. The leading term is the product of two genus-1 partition functions; the corrections are trace-class contributions from the sewing circle. The non-separating degeneration carries genuinely genus-2 information: the 2×2 period matrix controls the interplay of the two homology cycles.

Remark 4.9 (The Heisenberg as the atom of sewing). The Heisenberg algebra is the unique member of the standard landscape for which every step of analytic sewing is explicit: the sewing envelope is a classical Bergman space; the pair-of-pants amplitude is a Fock-space second quantisation; every partition function is a Fredholm determinant. All non-abelian chiral algebras satisfying HS-sewing require the abstract approach of Section 6. The lattice vertex algebra is the first extension (Section 5), where the Heisenberg Fredholm determinant appears as one factor of the full partition function.

4.8. **The Heisenberg at higher genus: Schottky uniformisation.** The Fredholm determinant formula for the Heisenberg partition function admits an interpretation in terms of Schottky uniformisation that illuminates the sewing structure.

A genus-g Schottky group is generated by g loxodromic Möbius transformations $\gamma_1,\ldots,\gamma_g \in \mathrm{PSL}_2(\mathbb{C})$, each with multiplier q_j ($|q_j| < 1$). The Schottky uniformisation represents Σ_g as a quotient of a domain in $\mathbb{CP}^1$ by the Schottky group.

The sewing construction realises this uniformisation at the chain level: each generator γ_j corresponds to a sewing circle with parameter q_j. Under the usual Schottky determinant normalisation for the free boson, the oscillator factor is expressed by a product over primitive conjugacy classes,

$$Z_g(\mathcal{H}^{(r)};\,q_1,\ldots,q_g) \;=\; \prod_{n\geq 1}\prod_{\{\gamma\}'}(1-q_\gamma^n)^{-r}, \tag{4.1}$$

where $\{\gamma\}'$ denotes the set of primitive conjugacy classes in the Schottky group and q_γ is the multiplier of γ. Absolute convergence is a theorem about the chosen Schottky domain and the growth of primitive conjugacy classes; it is not a consequence of a single multiplier inequality.

Remark 4.10 (Selberg zeta function). The Schottky product (4.1) is the Schottky analogue of the Selberg determinant expression for the scalar Laplacian. Identifying it with $\det'_\zeta \Delta_{\Sigma_g}$ requires the precise determinant normalisation and zero-mode convention. The paper uses only the trace-class sewing theorem above.

Remark 4.11 (The Polyakov conjecture). Polyakov's determinant formula computes the string partition function as a product of determinants: the holomorphic sector gives $(\det'_\zeta \Delta)^{-r/2}$ (the zeta-regularised determinant of the scalar Laplacian), and the non-holomorphic sector gives $(\det \operatorname{Im} \Omega)^{-r/2}$ (the volume of the Jacobian torus). The HS-sewing theorem provides the vertex-algebraic proof: the algebraic sewing (Fredholm determinant of the one-particle Bergman operator) reproduces the spectral-geometric answer (zeta-regularised determinant of the Laplacian), and both are equal to the Belavin–Knizhnik formula [2]. The three independent computations (algebraic, spectral, geometric) constitute a verification of the Heisenberg sewing theorem by independent paths.

5. Lattice sewing: theta functions

The lattice vertex algebra V_Λ associated to a positive-definite even lattice Λ of rank r provides the natural next case after the Heisenberg: analytic sewing combines the rank-r Heisenberg determinant with the convergence of the Siegel theta function.

5.1. Structure of the lattice vertex algebra.

Let Λ be a positive-definite even lattice of rank r. The lattice vertex algebra V_Λ is the tensor product of the rank-r Heisenberg Fock space with the group algebra of Λ:

$$V_\Lambda = \bigoplus_{\lambda \in \Lambda} \mathcal{F}_\lambda, \qquad \mathcal{F}_\lambda = \mathcal{H}_r \otimes \mathbb{C}e^\lambda.$$

The discriminant group Λ^*/Λ indexes simple coset modules of the full module category, not the charge summands of the vacuum VOA. The vertex operators e^γ ($\gamma \in \Lambda$) act by

$$e^\gamma \cdot a_{-n_1} \cdots a_{-n_j}|\lambda\rangle = \varepsilon(\gamma, \lambda) \cdot a_{-n_1} \cdots a_{-n_j}|\gamma + \lambda\rangle,$$

where $\varepsilon \colon \Lambda \times \Lambda \to \{\pm 1\}$ is the 2-cocycle. The central charge is $c = r$ and $\kappa(V_\Lambda) = r$.

Sector dimensions are the coefficients of $\Theta_\Lambda(q)/\eta(q)^r$. The sector growth is subexponential: $\log \dim(V_\Lambda)_n = O(\sqrt{n})$. OPE structure constants satisfy $|C^{c,k}_{a,i;b,j}| \le K(a+b+c+1)^r$ because all OPE singularities in V_Λ are at most r-th order poles.

5.2. The lattice partition function.

Theorem 5.1 (Lattice sewing). *Let Λ be a positive-definite even lattice of rank r.*

(i) ***(HS-sewing)*** *V_Λ satisfies HS-sewing at every $0 < q < 1$.*

(ii) ***(Factorisation)*** *The genus-g partition function factors as*

$$Z_g(V_\Lambda; \Omega) = \underbrace{(\det \operatorname{Im} \Omega)^{-r/2} \cdot (\det{}'_\zeta \Delta_{\Sigma_g})^{-r/2}}_{Z_g(\mathcal{H}_r; \Omega)} \cdot \underbrace{\sum_{\lambda \in \Lambda^g} e^{i\pi \lambda^T \Omega \lambda}}_{\Theta_\Lambda(\Omega)}, \tag{5.1}$$

and each factor converges absolutely on the Siegel upper half-space $\mathfrak{H}_g$.

(iii) ***(Sewing envelope)*** *The sewing envelope is the charge-refined Hilbert completion*

$$V_\Lambda^{\mathrm{sew}} = \widehat{\bigoplus}_{\lambda \in \Lambda} \operatorname{Sym} A^2_\lambda(D), \tag{5.2}$$

where $A^2_\lambda(D) \subset A^2(D)$ is the Bergman space with L_0-shift $\frac{1}{2}\langle \lambda, \lambda \rangle$ and the completion is in the HS-sewing topology.

(iv) ***(Charge shifts)*** *Vertex operators e^γ shift charge sectors by $\lambda \mapsto \lambda + \gamma$. They are bounded on fixed finite charge windows after the L_0-shift is included. No uniform boundedness over the full charge lattice is asserted.*

Proof. (i) follows from $\log \dim(V_\Lambda)_n = O(\sqrt{n})$ (subexponential sector growth) and $|C^{c,k}_{a,i;\,b,j}| \le K(a + b + c + 1)^r$ (polynomial OPE growth): the hypotheses of Theorem 6.1 are satisfied.

(ii): the Fock decomposition $V_\Lambda = \bigoplus_{\lambda\in\Lambda} \mathcal{F}_\lambda$ is a direct sum of Heisenberg Fock spaces indexed by lattice charges. Within each sector, Wick's theorem reduces all correlation functions to propagator pairings on the underlying rank-r Heisenberg, contributing $Z_g(\mathcal{H}_r;\ \Omega)$ (Theorem 4.4). The lattice sum over winding modes contributes $\Theta_\Lambda(\Omega)$. The Heisenberg factor converges by the one-particle sewing theorem (eigenvalues q^n, trace class). The theta factor converges absolutely on $\mathfrak{H}_g$: the defining sum satisfies $|\Theta_\Lambda(\Omega)| \le \sum_\lambda e^{-\pi\lambda^T Y\lambda}$ with $Y = \operatorname{Im}(\Omega) > 0$, and the exponent $\lambda^T Y\lambda \ge \lambda_{\min}(Y)\cdot\|\lambda\|^2$ grows as $\|\lambda\|^2 \to \infty$.

(iii): the Bergman-space completion $\widehat{\mathcal{F}}_\lambda = \overline{\mathcal{F}_\lambda}^{\|\cdot\|_{A^2}}$ gives honest Hilbert sectors, and the theta weight supplies the summable charge completion.

(iv): on a finite charge window, e^γ preserves oscillator mode operators and shifts only finitely many lattice vectors. The cocycle ε is bounded by 1, and the L_0-shift is bounded on that window. Uniform boundedness over all $\lambda \in \Lambda$ would require control of $\langle\gamma, \lambda\rangle$ and is not claimed. □

Remark 5.2 (Self-dual lattices and modular invariance). When Λ is unimodular ($D(\Lambda) = \{0\}$, e.g. E_8 or Λ_{24}), the theta function $\Theta_\Lambda(\Omega)$ is a Siegel modular form with respect to $\mathrm{Sp}_{2g}(\mathbb{Z})$. The full partition function $Z_g(V_\Lambda;\ \Omega)$ then transforms as a modular form of weight $r/2$, recovering Zhu's modular invariance theorem [10] from the sewing formalism. For non-self-dual lattices, the theta function transforms under a finite-index congruence subgroup of $\mathrm{Sp}_{2g}(\mathbb{Z})$, and the partition function has the corresponding transformation law.

Remark 5.3 (Genus-1 specialisation). At genus 1, the lattice partition function reduces to

$$Z_1(V_\Lambda;\ \tau) \;=\; \frac{\Theta_\Lambda(\tau)}{|\eta(\tau)|^{2r}}.$$

For the E_8 lattice ($r = 8$): the theta function $\Theta_{E_8}(\tau) = E_4(\tau)$ (the Eisenstein series of weight 4), and $Z_1(V_{E_8};\ \tau) = E_4(\tau)/|\eta(\tau)|^{16}$. For the Leech lattice ($r = 24$): $\Theta_{\Lambda_{24}}(\tau) = E_{12}(\tau) + \cdots$ (a weight-12 modular form).

Example 5.4 (Genus-2 for the E_8 lattice). At genus 2 with period matrix $\Omega = \begin{pmatrix}\tau_1 & z\\ z & \tau_2\end{pmatrix} \in \mathfrak{H}_2$, the E_8 lattice partition function is

$$Z_2(V_{E_8};\ \Omega) \;=\; Z_2(\mathcal{H}_8;\ \Omega)\cdot\Theta_{E_8}(\Omega),$$

where the Heisenberg factor is $Z_2(\mathcal{H}_8;\ \Omega) = (\det\operatorname{Im}\Omega)^{-4}\cdot(\det{}'_\zeta\Delta_{\Sigma_2})^{-4}$ and the genus-2 theta function is

$$\Theta_{E_8}(\Omega) \;=\; \sum_{\lambda\in E_8^2} e^{i\pi\lambda^T\Omega\lambda} \;=\; \sum_{\lambda_1,\lambda_2\in E_8} e^{i\pi(\tau_1|\lambda_1|^2 + 2z\langle\lambda_1,\lambda_2\rangle + \tau_2|\lambda_2|^2)}.$$

The genus-2 theta function $\Theta_{E_8}(\Omega)$ is a Siegel modular form of weight 4 with respect to $\mathrm{Sp}_4(\mathbb{Z})$. The factorisation of Z_2 into Heisenberg and lattice parts extends the genus-1 formula $Z_1 = E_4(\tau)/|\eta(\tau)|^{16}$ to genus 2.

The convergence of $\Theta_{E_8}(\Omega)$ on $\mathfrak{H}_2$ is controlled by the shortest vector of E_8 ($|\lambda_{\min}|^2 = 2$): the tail of the theta sum is bounded by $\sum_{|\lambda|\ge R} e^{-\pi\lambda_{\min}(Y)\cdot|\lambda|^2}$, which decays as $e^{-\pi\lambda_{\min}(Y)\cdot R^2}$.

Example 5.5 (The D_{16}^+ lattice and moonshine)**.** The lattice D_{16}^+ (rank 16, even, unimodular, but not isomorphic to $E_8 \oplus E_8$) provides a lattice VOA with the same genus-1 partition function as $V_{E_8\oplus E_8}$ (both equal $E_4(\tau)^2/|\eta(\tau)|^{32}$). The two VOAs are distinguished at genus 2: their Siegel theta functions $\Theta_{D_{16}^+}(\Omega) \neq \Theta_{E_8\oplus E_8}(\Omega)$ are distinct Siegel modular forms of weight 8. Analytic sewing detects this distinction through the genus-2 partition function, providing finer invariants than the genus-1 character.

5.3. **Convergence of the Siegel theta function.** The theta factor $\Theta_\Lambda(\Omega) = \sum_{\lambda\in\Lambda^g} e^{i\pi\lambda^T\Omega\lambda}$ in the lattice partition function requires a separate convergence analysis from the Heisenberg Fredholm factor. The two convergence mechanisms are logically independent.

Proposition 5.6 (Theta convergence on the Siegel upper half-space)**.** *Let Λ be a positive-definite even lattice of rank r. For any $\Omega \in \mathfrak{H}_g$ (i.e. $\Omega^T = \Omega$ and $Y := \operatorname{Im}\Omega > 0$), the theta series converges absolutely with the bound*

$$|\Theta_\Lambda(\Omega)| \;\leq\; \sum_{\lambda\in\Lambda^g} e^{-\pi\lambda^T Y\lambda} \;\leq\; C(Y)^{r/2}\cdot\big(\det Y\big)^{-r/2},$$

where $C(Y)$ is a constant depending on the lattice through the shortest vector.

Proof. For $\lambda = (\lambda_1,\dots,\lambda_g) \in \Lambda^g$, the real part of $i\pi\lambda^T\Omega\lambda$ is $-\pi\lambda^T Y\lambda$. Since $Y > 0$, each summand satisfies $|e^{i\pi\lambda^T\Omega\lambda}| = e^{-\pi\lambda^T Y\lambda}$. The quadratic form $\lambda^T Y\lambda \geq \lambda_{\min}(Y)\cdot\|\lambda\|^2$ grows as $\|\lambda\|^2$, so the sum converges by comparison with a Gaussian integral over $\mathbb{R}^{rg}$. The bound follows from the Poisson summation formula applied to the Gaussian $e^{-\pi\lambda^T Y\lambda}$. □

Remark 5.7 (Two independent convergence mechanisms). The lattice partition function (5.1) has two factors, each with a distinct convergence mechanism:

(a) the Heisenberg factor $Z_g(\mathcal{H}_r;\,\Omega)$ converges by the Fredholm determinant (trace-class sewing operator, eigenvalue decay q^n, $n \geq 1$);
(b) the theta factor $\Theta_\Lambda(\Omega)$ converges by positive-definiteness of $\operatorname{Im}\Omega$ (Gaussian decay of the lattice sum).

Neither mechanism alone suffices for the other factor. The Heisenberg factor involves the Bergman kernel and the Arakelov Green function; the theta factor involves the lattice geometry and the period matrix. The factorisation $Z_g = Z_g^{\mathrm{Heis}}\cdot\Theta_\Lambda$ is a consequence of the Fock decomposition of V_Λ into twisted Heisenberg modules; no analogous factorisation is available for non-abelian algebras.

5.4. **The Dirichlet–sewing lift for lattice VOAs.** The connected free energy of the lattice VOA at genus 1 is

$$F_1^{\mathrm{conn}}(V_\Lambda;\,q) \;=\; -r\log\det(1-T_q) - \log\Theta_\Lambda(\tau),$$

where the first term is the rank-r Heisenberg contribution and the second is the lattice correction. The Dirichlet–sewing lift produces

$$S_{V_\Lambda}(u) \;=\; r\,\zeta(u)\,\zeta(u+1) + S_\Lambda^{\mathrm{lattice}}(u),$$

where $S_\Lambda^{\mathrm{lattice}}(u)$ encodes the theta function's Fourier coefficients as a Dirichlet series. For self-dual lattices, the lattice Dirichlet series inherits the modular invariance of the theta function.

6. General HS-sewing theorem for the standard landscape

For chiral algebras beyond the Heisenberg and lattice families, the sewing envelope may lack a classical description. The Hilbert–Schmidt criterion can still be verified from two OPE-level estimates.

6.1. The general criterion.

Theorem 6.1 (General HS-sewing criterion). *Let $\mathcal{A}$ be a positive-energy chiral algebra with a choice of pre-Hilbert structure on each sector H_n. Suppose:*

(i) (**Hardy–Ramanujan sector growth**) $\dim H_n \le Ce^{\alpha\sqrt{n}}$ *for some* $C, \alpha > 0$*;*
(ii) (**polynomial OPE coefficient growth**) *there exist* $K > 0$ *and* $N \ge 0$ *such that the matrix elements of* $m^c_{a,b}$ *in orthonormal bases satisfy*

$$|C^{c,k}_{a,i;\,b,j}| \;\le\; K(a+b+c+1)^N$$

for all weights a, b, c *and all basis indices* i, j, k.

Then $\mathcal{A}$ satisfies HS-sewing at every parameter $0 < q < 1$.

Proof. From hypothesis (ii), the Hilbert–Schmidt norm of the sector multiplication is bounded by

$$\|m^c_{a,b}\|^2_{\mathrm{HS}} \;\le\; \dim H_a \cdot \dim H_b \cdot \dim H_c \cdot K^2(a+b+c+1)^{2N}. \tag{6.1}$$

The power-mean inequality $(x+y+z)^p \le 3^{p-1}(x^p+y^p+z^p)$ with $p = 2N$ gives

$$(a+b+c+1)^{2N} \;\le\; 3^{2N-1}\big((a+1)^{2N} + (b+1)^{2N} + (c+1)^{2N}\big).$$

Substituting into (6.1) and using the symmetry of the three terms:

$$\sum_{a,b,c\ge 0} q^{a+b+c}\,\|m^c_{a,b}\|^2_{\mathrm{HS}} \;\le\; 3^{2N}K^2\Big(\sum_{n\ge 0} q^n \dim H_n\,(n+1)^{2N}\Big)\Big(\sum_{n\ge 0} q^n \dim H_n\Big)^2.$$

By hypothesis (i), $\dim H_n \le Ce^{\alpha\sqrt{n}}$, while $q^n = e^{-n\log(1/q)}$ decays exponentially. Each factor $q^n \dim H_n\,(n+1)^{2N}$ is bounded by $C(n+1)^{2N}\exp(\alpha\sqrt{n} - n\log(1/q))$, which decays superpolynomially. Each series converges absolutely. □

Corollary 6.2 (Automatic HS-sewing for Koszul algebras). *Every finitely strongly generated chirally Koszul algebra satisfies HS-sewing after a positive Hilbert model is fixed and the PBW monomial basis has polynomial OPE matrix coefficients. Finite strong generation gives Hardy–Ramanujan sector growth; PBW gives the finite exponent in hypothesis (ii).*

6.2. **Verification on the standard landscape.** We verify the hypotheses of Theorem 6.1 for each standard family. The data required is a character formula (controlling $\dim H_n$) and a polynomial OPE bound (determining N).

Example 6.3 (Heisenberg). The Heisenberg $\mathcal{H}_k$ has $\dim H_n = p(n)$, the partition number. The OPE $J(z)J(w) \sim k/(z-w)^2$ has structure constants bounded by k, independent of weight; $N = 1$ suffices (the derivation ∂ increases weight by 1, so normal-ordered products of derivatives are at most linear in the weight).

Example 6.4 (Affine Kac–Moody). Let $V_k(\mathfrak{g})$ be the vacuum module at level k, with $\mathfrak{g}$ simple. Sectors satisfy $\dim H_n \le Ce^{\alpha\sqrt{n}}$ from the Kac–Weyl character formula. The OPE $J^a(z)J^b(w) \sim k\,\delta^{ab}/(z-w)^2 + f^{ab}{}_cJ^c(w)/(z-w)$ gives $|C| \le K(n+1)$ with $N = 1$. HS-sewing at every $0 < q < 1$ follows.

Example 6.5 (Virasoro). The Virasoro algebra Vir_c has $\dim H_n = p(n)$. The OPE $T(z)T(w) \sim (c/2)(z-w)^{-4} + 2T(w)(z-w)^{-2} + \partial T(w)(z-w)^{-1}$ is quartic (the r-matrix has cubic pole by d log-absorption, per the pole relation $\mathrm{pole}_r = \mathrm{pole}_{\mathrm{OPE}} - 1$). Repeated normal products give $|C| \le K(n+1)^2$ with $N = 2$ in the PBW monomial norm, and in the unitary range this is a genuine Hilbert estimate. At nonunitary or degenerate central charges the Shapovalov form is not a positive Hilbert form; the criterion applies only after choosing a positive quotient or completion for which the displayed polynomial bound is verified.

Example 6.6 ($\mathcal{W}_N$-algebras)**.** The principal $\mathcal{W}_N$-algebra has $N-1$ generators of conformal weights $2, 3, \ldots, N$. Sector dimensions satisfy $\dim H_n \le Ce^{\alpha\sqrt{n}}$ from the $\mathcal{W}_N$ character formula. OPE coefficient growth is $|C| \le K(n+1)^{2N}$ in the PBW monomial norm. Applying Theorem 6.1 gives HS-sewing at every $0 < q < 1$ in any positive Hilbert realization satisfying the same bound.

Example 6.7 (Lattice vertex algebras)**.** Covered by Theorem 5.1; recorded here for completeness. $\dim H_n = [q^n]\Theta_\Lambda(q)/\eta(q)^r$, $|C| \le K \cdot r$, $N = r$.

Proposition 6.8 (Summary table)**.** *The following table records the HS-sewing data for the standard landscape. The entries prove the criterion in the indicated positive Hilbert realizations; for nonunitary degenerate quotients the polynomial bound is an additional normed-model hypothesis.*

$\mathcal{A}$	$\dim H_n$	$\|C\|$ *bound*	N	$\sum_n q^n \dim H_n$
$\mathcal{H}_k$	$p(n)$	$\le k$	1	$q^{1/24}/\eta(q)$
$V_k(\mathfrak{g})$	$\le Ce^{\alpha\sqrt{n}}$	$\le K(n+1)$	1	$\chi_{V_k}(q)$
Vir_c	$p(n)$	$\le K(n+1)^2$	2	$q^{-c/24}\prod_{m\ge 2}(1-q^m)^{-1}$
$\mathcal{W}_N$	$\le Ce^{\alpha\sqrt{n}}$	$\le K(n+1)^{2N}$	$2N$	$\chi_{\mathcal{W}_N}(q)$
V_Λ	$[q^n]\Theta_\Lambda(q)/\eta(q)^r$	$\le Kr$	r	$\Theta_\Lambda(q)/\eta(q)^r$

Remark 6.9 (Completed algebras). The standard families above are finitely strongly generated. For completed algebras (such as $\mathcal{W}_{1+\infty}$ or completed Yangians), the polynomial OPE bound of hypothesis (ii) must be verified in the completed (inverse-limit) topology, where the structure constants are formal power series. This verification is the content of the MC4 completion theorem of [5].

Remark 6.10 (Beyond the standard landscape). The HS-sewing criterion is a priori applicable to any positive-energy chiral algebra satisfying subexponential growth and polynomial OPE bounds. Two classes of algebras beyond the standard landscape merit investigation:

(a) Admissible-level simple quotients $L_k(\mathfrak{g})$ satisfy C_2-cofiniteness [10], giving modular control of their characters. HS-sewing for such quotients still requires a positive Hilbert model and polynomial pair-of-pants matrix-coefficient bounds.

(b) $\mathcal{W}$-algebras at non-generic central charge, where the sector growth may exhibit resonance phenomena but remains subexponential.

6.3. **Fredholm expansion and the shadow tower.** The HS-sewing theorem gives trace-class control for closed stable sewing graphs. In Gaussian one-particle cases this is a Fredholm determinant; in general it is a trace-class stable-graph expansion.

For any HS-sewing algebra satisfying the bounded collar and heat-trace hypotheses, the genus-g partition function admits a controlled graph-sum expansion over stable graphs $\Gamma \in \mathcal{S}_g$ (the set of stable graphs of $\overline{\mathcal{M}}_g$):

$$Z_g(\mathcal{A};\Omega) = \sum_{\Gamma\in\mathcal{S}_g} \frac{1}{|\mathrm{Aut}(\Gamma)|} \prod_{v\in V(\Gamma)} A_v(\mathcal{A}) \prod_{e\in E(\Gamma)} K_e(q_e), \tag{6.2}$$

where A_v is the vertex amplitude (genus-g_v with n_v legs) and $K_e(q_e)$ is the sewing kernel along edge e. Each controlled term is trace class by Proposition 3.2.

The shadow obstruction tower $\Theta_{\mathcal{A}} = \sum_{r\ge 2} S_r(\mathcal{A})$ governs the formal leading terms of the graph-sum. At uniform weight, the genus-g partition function satisfies $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT), where λ_g^{FP} is the Faber–Pandharipande evaluation of the tautological class λ_g on $\overline{\mathcal{M}}_g$.

Proposition 6.11 (Controlled shadow evaluations)**.** *Under HS-sewing together with the bounded collar, heat-trace, and finite stable-graph hypotheses above, every fixed finite shadow evaluation that is represented by those controlled sewing graphs converges to a finite complex number.*

Proof. Each fixed finite evaluation is a finite linear combination of the controlled graph amplitudes in (6.2). The trace-class property gives convergence term by term. HS-sewing alone does not prove convergence of arbitrary tautological evaluations or of an infinite sum over all shadow degrees. ☐

6.4. **The Dirichlet–sewing lift.** The connected free energy at genus 1 encodes connected sewing amplitudes $a_{\mathcal{A}}(N)$ as Fourier coefficients. The Dirichlet–sewing lift

$$S_{\mathcal{A}}(u) \;=\; \sum_{N\geq 1} a_{\mathcal{A}}(N)N^{-u} \;=\; \frac{(2\pi)^u}{\Gamma(u)}\int_0^\infty F_{\mathcal{A}}^{\mathrm{conn}}(e^{-2\pi y})\, y^{u-1}\,dy$$

is a Dirichlet series whose analytic structure is controlled by the MC element $\Theta_{\mathcal{A}}$.

Example 6.12 (Explicit Dirichlet series)**.** The sewing Dirichlet series for the standard families:

$$\begin{aligned}
S_{\mathcal{H}_k}(u) &= k\,\zeta(u)\,\zeta(u+1),\\
S_{V_k(\mathfrak{g})}(u) &= \dim(\mathfrak{g})\cdot\zeta(u)\,\zeta(u+1),\\
S_{\mathrm{Vir}_c}(u) &= \zeta(u+1)\big(\zeta(u)-1\big),\\
S_{\mathcal{W}_N}(u) &= \zeta(u+1)\Big((N-1)\zeta(u)-\textstyle\sum_{j=1}^{N-1}H_j(u)\Big),
\end{aligned}$$

where $H_j(u) = \sum_{n=1}^{j} n^{-u}$ are partial harmonic sums. The Heisenberg and Kac–Moody series have a double pole at $u=0$; the Virasoro series has a simple pole (the subtraction of 1 from $\zeta(u)$ removes the $N=1$ term).

The sewing amplitudes $a_{\mathcal{A}}(N)$ define a positive-semidefinite inner product $\langle f,g\rangle_{\mathcal{A}} = \sum_N a_{\mathcal{A}}(N)f(N)g(N)$ with reproducing kernel $K_{\mathcal{A}}(s,t) = S_{\mathcal{A}}(s+t)$. The surface moment matrix $\mathscr{M}_{ij} = K_{\mathcal{A}}(\alpha/2+i,\alpha/2+j)$ is positive definite for all $\alpha\geq 2$: a Gram matrix V^TDV with Vandermonde V.

7. The three-layer analytic gap

The HS-sewing theorem guarantees convergent partition functions for the standard landscape, but three layers of structure separate the convergence statement from a complete analytic theory.

7.1. **Layer 1: the sewing envelope for interacting algebras.** For the Heisenberg algebra, the sewing envelope is $\mathrm{Sym}A^2(D)$: a classical Fock space built on the Bergman space of the disk. For lattice vertex algebras, it is the charge-refined Hilbert completion (5.2). Both identifications rest on the same mechanism: the Wick theorem reduces all amplitudes to one-particle (Heisenberg) or one-particle-plus-lattice (lattice) data, and the sewing envelope inherits the Bergman structure from the one-particle space.

For non-abelian chiral algebras (affine Kac–Moody at generic level, Virasoro, $\mathcal{W}_N$), no such reduction is available. The sewing envelope exists abstractly by Definition 2.1. Under the controlled closed graph hypotheses, HS-sewing makes the corresponding weighted completion H_q a concrete Hilbert space on which those closed amplitudes are trace class. A structural identification of $\mathcal{A}^{\mathrm{sew}}$ analogous to the Bergman identification for the Heisenberg is a separate problem.

Conjecture 7.1 (Sewing envelope for $V_k(\mathfrak{g})$)**.** *The sewing envelope of $V_k(\mathfrak{g})$ at integer level $k\in\mathbb{Z}_{>0}$ is the ind-Hilbert completion $\widehat{V_k(\mathfrak{g})}^{\mathrm{IndHilb}}$, and the controlled closed amplitudes admit trace-class stable-graph expansions indexed by Weyl-orbit data.*

The next step is explicit: the case $\mathfrak{g} = \mathfrak{sl}_2$ at $k = 1$ is the simplest interacting sewing computation.

Example 7.2 ($\mathfrak{sl}_2$ at level 1)**.** The vacuum module $V_1(\mathfrak{sl}_2)$ has central charge $c = 1$ and $\kappa = \dim(\mathfrak{sl}_2)(1+2)/(2 \cdot 2) = 3 \cdot 3/4 = 9/4$. At conformal weight 2, the sector H_2 is 3-dimensional, spanned by $\{J^a_{-1}J^b_{-1}\mathbf{1}\}_{a,b\in\{+,-,0\}}$ modulo the Serre relations. The pair-of-pants sewing kernel restricted to this sector is a 3×3 matrix whose entries are determined by the fusion rules $V_1 \otimes V_1 \to V_1$ and the KZ connection at level 1.

The KZ equation at level 1 is $\partial_z\Phi = \frac{\Omega}{(1+2)z}\Phi = \frac{\Omega}{3z}\Phi$ (KZ normalisation with Sugawara denominator $k + h^\vee = 3$), where Ω is the Casimir in $\mathfrak{sl}_2 \otimes \mathfrak{sl}_2$. The eigenvalues of the sewing kernel K_1 on the 3-dimensional sector are $\{q, q^2, q^2\}$ (the level-1 WZW primary fields have conformal weights 0 and $1/4$), so the trace-class condition is satisfied with $\|K_1|_{H_2}\|_1 = q + 2q^2$.

This is the gateway to a full interacting sewing computation: the first case where the sewing kernel is a non-diagonal matrix and the Fredholm expansion involves genuine fusion data.

Remark 7.3 (The conformal-block obstruction)*.* For non-abelian chiral algebras at generic (non-integer) level, the spaces of conformal blocks are infinite-dimensional; the sewing kernel is an infinite-rank operator on these spaces. HS-sewing guarantees that this operator is trace class, but the explicit spectral decomposition requires detailed knowledge of the conformal blocks (solutions of the KZ equation). For integer level, the conformal blocks are finite-dimensional (Zhu's theorem), and the sewing kernel is a finite matrix at each degree; this is the regime where explicit computation is feasible.

7.2. **Layer 2: metric independence of the ind-Hilbert factorisation.** Moriwaki's ind-Hilbert completion [6] is defined using a conformal metric on the curve; the Bergman inner product on $A^2(D)$ depends on the disk D and hence on the conformal structure. The algebraic bar complex, by contrast, depends only on the OPE data and is independent of any metric.

Definition 7.4 (Metric independence)**.** An ind-Hilbert factorisation is *metrically independent* if the isomorphism class of the completed sewing envelope (as a topological vector space with continuous pair-of-pants multiplication) is independent of the conformal metric used to define the Bergman inner product.

For the Heisenberg, metric independence is immediate: the Fock space $\mathrm{Sym}A^2(D)$ depends on the disk D, but any two disks are conformally equivalent, and the isomorphism class of the Bergman space is a conformal invariant.

For interacting algebras, metric independence is the assertion that the analytic completion sees only the algebraic structure (the OPE), not the auxiliary metric. This is a functional-analytic statement with no algebraic counterpart; it is the content of the algebraic-to-analytic comparison theorem.

Conjecture 7.5 (Metric independence)**.** *For every positive-energy chiral algebra satisfying HS-sewing, the ind-Hilbert factorisation is metrically independent: the sewing envelope $\mathcal{A}^{\mathrm{sew}}$ depends only on the algebraic chiral core $\mathcal{A}_{\mathrm{alg}}$.*

Remark 7.6 (Evidence for metric independence)*.* Three observations support the conjecture:

(a) For the Heisenberg, the Bergman space $A^2(D)$ depends on D, but the abstract Hilbert space structure (separable, infinite-dimensional, with the canonical grading by polynomial degree) is a conformal invariant. Any two conformal metrics give isomorphic Bergman spaces; the isomorphism is the conformal change-of-variables on holomorphic functions.

(b) The algebraic bar differential depends only on OPE data (residues at collision diagonals); the OPE coefficients are independent of conformal metric. If the bar differential is dense in the completed bar differential (as guaranteed by the analytic-algebraic comparison), the topology of the completion is determined by the algebraic data.

(c) The sewing envelope is defined by seminorms (2.1) that depend on bordism amplitudes. Each amplitude is a composition of OPE operations (metric-independent) and sewing traces (metric-dependent only through the choice of conformal collar). The collar dependence enters through the sewing parameter q, which appears universally in all completions.

The conjecture is that these observations extend to a proof for all HS-sewing algebras: the sewing topology depends on q but not on the conformal metric that determines the embedding of the collar in the surface.

7.3. **Layer 3: the coderived shadow at $g \geq 1$.** At genus $g = 0$ the bar complex satisfies $d^2 = 0$, and the ordinary derived category is the correct receptacle. At genus $g \geq 1$, the fiberwise bar differential acquires curvature:

$$d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g,$$

where $\kappa(\mathcal{A})$ is the modular characteristic and $\omega_g = c_1(\lambda)$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_g$. For $\kappa \neq 0$ (which holds for every non-trivial chiral algebra in the standard landscape), $d_{\text{fib}}^2 \neq 0$ and ordinary cohomology of the fiberwise complex is not defined.

The correct receptacle is the coderived category D^{co}, developed in the next section. The passage from derived to coderived is the categorified form of passage from the equation $d^2 = 0$ to the curved Maurer–Cartan equation $d\alpha + \alpha^2 + \kappa\omega = 0$.

The analytic content of Layer 3 is: the sewing amplitudes that converge under the controlled sewing hypotheses must be reinterpreted as evaluations of coderived (not derived) functors. The Fredholm determinant of the Heisenberg is the prototype: the curvature $d_{\text{fib}}^2 = k \cdot \omega_1$ at genus 1 means that the bar complex does not define an object in the derived category; the Fredholm determinant $\det(1 - T_q) = q^{-1/24}\eta(q)$ is the correct coacyclic invariant.

7.4. **Interaction between the three layers.** The three layers are logically independent but interact in concrete examples.

Proposition 7.7 (Layer independence)**.**

(i) *Layer 1 (sewing envelope identification) does not imply Layer 2 (metric independence): an explicit Hilbert identification could depend on the choice of conformal metric.*

(ii) *Layer 2 (metric independence) does not imply Layer 3 (coderived formulation): metric independence is a genus-*0 *property of the completion, while the coderived category is forced by curvature at* $g \geq 1$.

(iii) *Layer 3 (coderived formulation) does not imply Layer 1: the coderived category provides the correct homotopy theory but does not identify the completion concretely.*

Proof. (i): the Heisenberg sewing envelope $\mathrm{Sym}\,A^2(D)$ depends on the disk D; the metric independence is a separate theorem (conformal invariance of the Bergman space, Proposition A.6).

(ii): metric independence concerns the topology of the completion; the coderived category concerns the homological algebra of the curved differential. The Heisenberg at genus 0 is metrically independent but has $d^2 = 0$ (no coderived issues); at genus 1 the coderived formulation is needed but the metric independence is inherited from genus 0.

(iii): the coderived category provides $C_{\text{BV}}^\bullet \simeq_{D^{\text{co}}} B^{(g)}$ abstractly; the identification of $\mathcal{A}^{\text{sew}}$ with a specific Hilbert space is additional data. □

For the standard landscape, the abelian families supply the explicit sewing envelope and metric-independence models. The non-abelian families require positive Hilbert realisations with polynomial matrix-coefficient bounds. The coderived comparison is the homological target once the stated coacyclicity hypotheses are available.

7.5. **The sewing envelope as a nuclear space.** The weighted spaces H_q form a natural nuclear Hilbert scale as $q \to 1^-$. Identifying this scale with the sewing envelope requires the sewing topology and the weighted Hilbert-scale topology to control one another; HS-sewing alone gives only one comparison map.

Proposition 7.8 (Nuclear Hilbert scale). *Assume the sector dimensions have subexponential growth. Then the weighted completions form a projective Hilbert scale*

$$\mathcal{H}^{\mathrm{scale}} := \varprojlim_{q \to 1^-} H_q$$

whose transition maps $H_{q'} \to H_q$ (for $q' > q$) are Hilbert–Schmidt. In particular, $\mathcal{H}^{\mathrm{scale}}$ is a nuclear Fréchet space. If, in addition, the sewing seminorms and the Hilbert-scale seminorms define the same completion on $\mathcal{A}_{\mathrm{alg}}$, then $\mathcal{A}^{\mathrm{sew}} \cong \mathcal{H}^{\mathrm{scale}}$.

Proof. The weighted completions H_q form a projective system: for $q' > q$, the inclusion $\mathcal{A}_{\mathrm{alg}} \hookrightarrow H_q$ factors through $H_{q'}$ because $q'^n > q^n$ for all $n \geq 1$. The transition map $H_{q'} \to H_q$ sends the basis vector $e_{n,j} \in H_n$ to $(q/q')^n e_{n,j}$, with Hilbert–Schmidt norm $\sum_{n,j} (q/q')^{2n} = \sum_n \dim H_n (q/q')^{2n}$, which is finite by subexponential sector growth (the ratio $q/q' < 1$ provides exponential decay). A projective limit of Hilbert spaces with Hilbert–Schmidt transition maps is nuclear [8]. □

The nuclear structure implies the Schwartz kernel theorem on $\mathcal{H}^{\mathrm{scale}}$. It applies to $\mathcal{A}^{\mathrm{sew}}$ only after the additional topology-comparison hypothesis in Proposition 7.8.

8. The coderived category as correct receptacle

The genus $g \geq 1$ bar complex carries curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$. Ordinary cohomology is not the right invariant for such objects: if $d^2 \neq 0$, then $\operatorname{im} d$ need not lie in $\ker d$. The coderived category D^{co} supplies the homotopy category in which curved differentials and coacyclic cones are handled directly.

8.1. Curved dg-modules.

Definition 8.1 (Curved dg-module). A *curved dg-module* over a graded algebra R is a graded R-module M equipped with a degree-1 endomorphism d satisfying

$$d^2 = \mu \cdot \mathrm{id}_M$$

for some central element $\mu \in R$ (the *curvature*). When $\mu = 0$ this reduces to an ordinary dg-module.

The genus-g bar complex $B^{(g)}(\mathcal{A})$ is a curved dg-module over $R = H^*(\overline{\mathcal{M}}_g)$ with curvature $\mu = \kappa(\mathcal{A}) \cdot \omega_g$.

Warning 8.2 (Ordinary cohomology does not apply). If $d^2 = \mu \neq 0$, the usual cohomology quotient $\ker d / \operatorname{im} d$ is not defined in general, because $d(dx) = \mu x$ need not vanish. One must work with the curved object and its curved homotopy theory rather than forcing an ordinary derived category.

8.2. The coderived category.

The coderived category replaces acyclicity with coacyclicity.

Definition 8.3 (Coacyclic complex). A curved dg-module M is *coacyclic* if it lies in the smallest thick subcategory of the homotopy category of curved dg-modules containing all totalizations of short exact sequences. Equivalently, M is coacyclic if every morphism from M to an injective curved dg-module is null-homotopic.

Definition 8.4 (Coderived category). The *coderived category* $D^{\mathrm{co}}(R\text{-mod}_\mu)$ is the Verdier quotient of the homotopy category of curved dg-modules by the thick subcategory of coacyclic complexes.

The coderived category is the homotopy category used here for curved complexes: it retains the curved differential and quotients by coacyclic objects.

Remark 8.5 (Positselski's theory). Positselski [7] introduced the coderived category for curved dg-algebras and matrix factorisations. The totalisation of a short exact sequence of curved complexes need not be acyclic, since $d^2 \neq 0$, but it is always coacyclic. Coacyclicity is the correct curved replacement for ordinary acyclicity.

8.3. **The curvature bicomplex.** The curved bar complex $B^{(g)}(\mathcal{A})$ carries a natural bicomplex structure. Write the total differential as

$$D_g \;=\; d_{\mathrm{fib}} + \nabla_g^{\mathrm{GM}},$$

where d_{fib} is the fiberwise bar differential (with curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$) and ∇_g^{GM} is the Gauss–Manin connection. The total differential satisfies

$$D_g^2 \;=\; d_{\mathrm{fib}}^2 + [d_{\mathrm{fib}}, \nabla_g^{\mathrm{GM}}] + (\nabla_g^{\mathrm{GM}})^2 \;=\; \kappa\omega_g - \kappa\omega_g \;=\; 0: \tag{8.1}$$

flatness is restored by coupling the bar differential to the Hodge bundle. The curvature bicomplex is the bigraded object with horizontal differential d_{fib} and vertical differential ∇_g^{GM}.

Proposition 8.6 (Totalisation in D^{co})**.** *The totalisation of the curvature bicomplex of $B^{(g)}(\mathcal{A})$ is an object of the coderived category D^{co}. When the controlled closed sewing hypotheses of Proposition 3.2 hold, the associated closed amplitude is the trace of the totalised sewing operator.*

Proof. The totalisation of the bicomplex $(d_{\mathrm{fib}}, \nabla_g^{\mathrm{GM}})$ produces a complex with $D_g^2 = 0$ by (8.1), hence an honest object of the derived category of the total complex. The passage to the coderived category of the curved component d_{fib} (with $d_{\mathrm{fib}}^2 = \kappa\omega_g$) is by the standard Positselski comparison: totalisation of the bicomplex represents the curved component as an object of D^{co} because the vertical differential ∇_g^{GM} absorbs the curvature.

Under the controlled closed sewing hypotheses, the trace-class property of Proposition 3.2 ensures that the trace of the totalised sewing operator is an absolutely convergent number. In Gaussian one-particle cases this trace can be rewritten as a Fredholm determinant; in general it remains a trace-class stable-graph trace. □

8.4. **The BV/bar comparison in the coderived category.** The BV complex of a chiral algebra on a Riemann surface Σ_g is the Costello–Gwilliam perturbative quantisation [3]; the bar complex $B^{(g)}(\mathcal{A})$ is the algebraic chain model. At genus 0, the two are chain-level quasi-isomorphic. At genus $g \geq 1$, curvature intervenes: the BV propagator acquires a harmonic component (the Arakelov Green function) that produces corrections invisible to the algebraic bar differential.

Proposition 8.7 (Conditional BV/bar comparison in the coderived category)**.** *Let $\mathcal{A}$ be a chirally Koszul algebra for which the curved factorization model supplies an m_0-Koszul totalization with coacyclic comparison cones. For each genus $g \geq 1$, let*

$$f_g\colon C^\bullet_{\mathrm{BV}}(\mathcal{A}, \Sigma_g) \longrightarrow B^{(g)}(\mathcal{A})$$

be the comparison morphism obtained by replacing the BV propagator with its Hodge decomposition relative to the bar propagator. If the cone of f_g is coacyclic, then

$$C^\bullet_{\mathrm{BV}}(\mathcal{A}, \Sigma_g) \;\simeq_{D^{\mathrm{co}}}\; B^{(g)}(\mathcal{A}).$$

For class M*, coacyclicity requires an explicit proof that the harmonic discrepancy is curvature-divisible.*

Proof. At genus 0, the comparison is the chain-level quasi-isomorphism: $d_{\mathrm{fib}}^2 = 0$ and the harmonic component of the propagator vanishes on $\mathbb{C}$.

For $g \geq 1$, Hodge decomposition writes the BV propagator as the sum of the bar propagator (holomorphic, $d \log E(z,w)$, weight 1) and the harmonic correction $h_g = (2\pi)^{-1}(\mathrm{Im}\,\Omega)^{-1}_{ab}\, d\bar{z}_a \wedge dz_b$. The harmonic correction produces a discrepancy between the BV and bar differentials.

The proposition is formal after the coacyclic-cone hypothesis: a morphism with coacyclic cone becomes an isomorphism in the Verdier quotient defining D^{co}. The class-by-class statements are proof obligations for that hypothesis. For class M this means computing the harmonic discrepancy and proving its curvature-divisibility; the present proposition does not supply that calculation. □

Remark 8.8 (Class-by-class obstruction profile). The four shadow classes suggest the following obstruction profile:

Class	Chain-level status	Mechanism
G (Heisenberg)	weak equivalence	no interaction vertices
L (affine KM)	weak equivalence	Jacobi kills cubic harmonic term
C ($\beta\gamma$)	conditional	composite free-field factorisation
M (Virasoro, $\mathcal{W}_N$)	coderived only	harmonic correction curvature-divisible

The table records the expected mechanism to check in each family. Only rows whose coacyclic-cone hypothesis has been verified in a chosen model give a theorem.

8.5. **The harmonic discrepancy mechanism.** The key to the coderived comparison is the factorisation of the harmonic discrepancy through curvature powers. The mechanism is the Hodge decomposition of the Bergman kernel.

On a genus-g surface Σ_g with period matrix Ω, the Bergman kernel (the bar propagator) is

$$B(z,w) \;=\; d_z d_w \log E(z,w) \;+\; 2\pi i\,(\mathrm{Im}\,\Omega)^{-1}_{ab}\,\omega_a(z)\,\omega_b(w),$$

where $E(z,w)$ is the prime form and $\omega_1, \ldots, \omega_g$ is a normalised basis of abelian differentials. The second term is the harmonic correction.

The BV propagator P_g differs from the bar propagator $d \log E(z,w)$ by this harmonic term:

$$P_g(z,w) \;=\; d \log E(z,w) \;+\; h_g(z,w), \qquad h_g \;=\; \frac{1}{2\pi}\,(\mathrm{Im}\,\Omega)^{-1}_{ab}\, d\bar{z}_a \wedge dz_b.$$

Each insertion of h_g into a Feynman graph vertex produces a harmonic correction to the BV amplitude. At the quartic level ($r = 4$), the correction is

$$\delta_4^{\mathrm{harm}} \;=\; Q^{\mathrm{contact}}(\mathcal{A}) \cdot \frac{\kappa(\mathcal{A})}{\mathrm{Im}(\tau)},$$

where Q^{contact} is the quartic contact invariant of $\mathcal{A}$. For class G and L, $Q^{\mathrm{contact}} = 0$. For class M, $Q^{\mathrm{contact}} \neq 0$ and δ_4^{harm} is a genuine obstruction to the chain-level comparison.

The crucial observation is that every δ_r^{harm} is divisible by $m_0 = \kappa \cdot \omega_g$ (the curvature element of the curved bar differential). This divisibility is the mechanism by which the comparison cone becomes coacyclic: in the coderived category, elements divisible by the curvature are trivial.

8.6. **Coderived versus derived: a comparison.** The passage from derived to coderived captures a precise mathematical phenomenon.

	Derived $D(R)$	**Coderived** $D^{co}(R)$
Curvature	$d^2 = 0$ required	$d^2 = \mu$ permitted
Acyclicity notion	$H^*(M) = 0$	coacyclic
Genus-0 bar	object of $D(R)$	object of $D^{co}(R)$
Genus-g bar ($\kappa \neq 0$)	empty	well-defined
BV/bar comparison	chain-level (G,L)	all classes
Partition function	formal series	convergent number

The coderived category is the categorified avatar of the curved Maurer–Cartan equation. The passage from $d^2 = 0$ to $d\alpha + \alpha^2 + \kappa\omega = 0$ is the passage from derived to coderived: the curvature becomes structural data in the differential.

8.7. **Genus-1 illustration.** The genus-1 case illustrates the three categories concretely.

At genus 1, the bar complex $B^{(1)}(\mathcal{A})$ has curvature $d^2_{\text{fib}} = \kappa(\mathcal{A}) \cdot \omega_1$, where $\omega_1 = c_1(\lambda)$ is the first Chern class of the Hodge bundle on $\overline{\mathcal{M}}_{1,1}$. The total differential $D_1 = d_{\text{fib}} + \nabla^{\text{GM}}$ satisfies $D_1^2 = 0$.

Example 8.9 (Heisenberg at genus 1)**.** For $\mathcal{H}_k$ with $\kappa = k$:

(a) *Derived category:* $d^2_{\text{fib}} = k \cdot \omega_1 \neq 0$ for $k \neq 0$. The fiberwise curved object is not an ordinary cochain complex, so $H^*(B^{(1)}(\mathcal{H}_k), d_{\text{fib}})$ is not the relevant invariant.

(b) *Coderived category:* the totalisation of the bicomplex $(d_{\text{fib}}, \nabla^{\text{GM}})$ is a well-defined object of D^{co}. For a rank-r Heisenberg VOA, the trace of the sewing operator gives $Z_1 = (\text{Im}\,\tau)^{-r/2}\,|\eta(\tau)|^{-2r} \neq 0$.

(c) *Total category:* $D_1^2 = 0$; the total complex $B^{(1)}(\mathcal{H}_k)$ is an honest dg-module over $H^*(\overline{\mathcal{M}}_{1,1})$ and lives in the ordinary derived category of the total complex.

The partition function Z_1 is a coderived invariant (well-defined in D^{co} of the curved fiberwise complex) and simultaneously a derived invariant (well-defined in D of the total complex). The two perspectives are related by the totalisation functor of Proposition 8.6.

Example 8.10 (Virasoro at genus 1)**.** For Vir_c with $\kappa = c/2$:

(a) The curvature is $d^2_{\text{fib}} = (c/2) \cdot \omega_1$.

(b) The partition function is $Z_1(\text{Vir}_c;\, \tau) = (\text{Im}\,\tau)^{-c/4}\,|\eta(\tau)|^{-2(c-1)}$ (the vacuum character of the Virasoro algebra).

(c) The BV/bar comparison: the harmonic discrepancy at the quartic level is $\delta_4^{\text{harm}} = Q^{\text{contact}} \cdot (c/2)/\text{Im}(\tau)$. For $c \neq 0$ (non-trivial Virasoro), this is nonzero; the chain-level BV/bar comparison fails. The cone is coacyclic by Theorem 8.7(iv), so the coderived comparison holds.

The Virasoro at genus 1 is the simplest class-M example: the harmonic discrepancy is explicit, the coacyclicity is verifiable, and the coderived category is genuinely necessary.

8.8. **Convergent versus formal partition functions.** The three levels of structure (algebraic, coderived, analytic) produce three notions of partition function:

(i) *Formal:* the genus-g component of the MC element $\Theta_{\mathcal{A}}$, evaluated on the fundamental class $[\overline{\mathcal{M}}_g]$. This is a formal power series in the sewing parameters $q_1, \ldots, q_g$. It exists for every chiral algebra with $D^2 = 0$.

(ii) *Coderived:* the trace of the totalised sewing operator in the coderived category. This is a formal power series with the additional property that each coefficient is a coderived invariant (independent of the choice of resolution up to coacyclic error). It exists for every positive-energy chiral algebra.

(iii) *Analytic:* an absolutely convergent complex number, the trace of a trace-class operator on the weighted Hilbert completion H_q. This exists under HS-sewing.

Proposition 8.11 (Compatibility of the three notions)**.** *For a chirally Koszul algebra satisfying HS-sewing:*

(a) *the formal partition function is the Taylor expansion of the analytic partition function around the cusp $q_j \to 0$;*
(b) *the coderived partition function is the image of the analytic partition function under the forgetful functor from the Hilbert-completed category to the coderived category;*
(c) *all three notions agree on the domain of convergence.*

Proof. (a): the formal MC element $\Theta_{\mathcal{A}}$ is the generating function for the graph-sum amplitudes, which are the Taylor coefficients of the partition function at the cusp. Under HS-sewing, the Taylor series converges absolutely for $|q_j| < 1$ (Corollary 3.3), so the formal and analytic partition functions agree on the domain of convergence.

(b): the forgetful functor sends a trace-class operator on H_q to its underlying curved dg-module structure in D^{co}; the trace (a convergent number) is sent to the same number viewed as a coderived invariant.

(c): on the domain of convergence, all three are the same complex number, computed by the same graph-sum with the same absolutely convergent Fredholm expansion. □

8.9. **The analytic Koszul pair.** The algebraic Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ controls Theorems A through H of modular Koszul duality at the algebraic level. The analytic extension is the following.

Definition 8.12 (Analytic Koszul pair)**.** An *analytic Koszul pair* is a pair $(\mathcal{A}, \mathcal{A}^!)^{\mathrm{an}}$ of chirally Koszul algebras, both satisfying HS-sewing, for which the analytic bar-cobar adjunction restricts to a Quillen equivalence on sewing-complete categories:

$$\bar{B}^{\mathrm{an}} : \mathcal{A}^{\mathrm{sew}}\text{-mod} \rightleftarrows (\mathcal{A}^!)^{\mathrm{sew}}\text{-comod} : \Omega^{\mathrm{an}}.$$

Every algebraic Koszul pair in the standard landscape is expected to extend to an analytic Koszul pair; the HS-sewing theorem provides the analytic input for both $\mathcal{A}$ and $\mathcal{A}^!$.

Conjecture 8.13 (Analytic realisation criterion)**.** *HS-sewing on $\mathcal{A}$ plus dual HS-sewing on $\mathcal{A}^!$ suffices for the analytic Koszul pair to exist: the analytic bar-cobar adjunction restricts to a Quillen equivalence on sewing-complete categories.*

The verification of this conjecture for the Heisenberg ($\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*) \neq \mathcal{H}_{-k}$) and for lattice vertex algebras is the natural test case.

Remark 8.14 (Koszul complementarity and the analytic pair)*.* The Koszul complementarity theorem (Theorem C of the modular Koszul duality theorem) asserts that $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = \varkappa(\mathcal{A})$ (the scalar complementarity invariant): $\varkappa = 0$ for KM/Heis/lattice/free; $\varkappa = 13$ for Virasoro; $\varkappa = 250/3$ for $\mathcal{W}_3$. The scalar $\varkappa$ is distinct from the Trinity ghost-charge conductor $K(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^!) = -c_{\mathrm{ghost}}(\mathrm{BRST})$ (taking values $K(\mathrm{Vir}) = 26$, $K(\mathcal{W}_3) = 100$, $K(\mathrm{BP}) = 196$); the two invariants are related by $\varkappa = \varrho_{\mathcal{A}} \cdot K$ with ϱ family-specific. The analytic content is: both $\mathcal{A}$ and $\mathcal{A}^!$ satisfy HS-sewing (because both are in the standard landscape), and the complementarity relation constrains the sewing data:

$$S_{\mathcal{A}}(u) + S_{\mathcal{A}^!}(u) \;=\; \varkappa \cdot \zeta(u)\, \zeta(u+1) \;+\; \text{lower-order terms},$$

where the leading term involves the scalar complementarity invariant. For the Virasoro pair $(\mathrm{Vir}_c, \mathrm{Vir}_{26-c})$: $S_{\mathrm{Vir}_c}(u) + S_{\mathrm{Vir}_{26-c}}(u) = 2\zeta(u+1)(\zeta(u) - 1)$, independent of c; the dependence on c cancels in the sum, reflecting the topological nature of the Koszul conductor.

Remark 8.15 (The analytic bar coalgebra). The algebraic bar coalgebra $B(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ carries the deconcatenation coproduct $\Delta\colon B(\mathcal{A}) \to B(\mathcal{A}) \otimes B(\mathcal{A})$. Under HS-sewing, the bar coalgebra completes to the *analytic bar coalgebra* $B^{\mathrm{an}}(\mathcal{A})$, defined as the graph-norm closure of $B(\mathcal{A})$ in the Hilbert-space topology induced by H_q. The coproduct extends continuously: $\Delta^{\mathrm{an}}\colon B^{\mathrm{an}}(\mathcal{A}) \to B^{\mathrm{an}}(\mathcal{A})\widehat{\otimes}B^{\mathrm{an}}(\mathcal{A})$, where $\widehat{\otimes}$ is the completed Hilbert-space tensor product.

The analytic bar coalgebra is the correct object for the analytic bar-cobar adjunction: it carries both the algebraic coalgebra structure (deconcatenation) and the analytic topology (trace-class sewing), and the cobar functor Ω^{an} applied to $B^{\mathrm{an}}(\mathcal{A})$ recovers the sewing envelope $\mathcal{A}^{\mathrm{sew}}$.

8.10. **The curvature–sewing interplay.** The coderived category and the HS-sewing condition interact at genus $g \geq 1$ through the curvature.

Proposition 8.16 (Curvature and trace class). *Let $\mathcal{A}$ satisfy HS-sewing and let $\kappa = \kappa(\mathcal{A})$. At genus $g \geq 1$, the curvature element $m_0 = \kappa \cdot \omega_g$ acts on the weighted completion H_q as a bounded operator of norm $\|m_0\|_{\mathrm{op}} = |\kappa| \cdot \|\omega_g\|$, where $\|\omega_g\|$ is the sup-norm of the Hodge class representative on the moduli space. The iterated curvatures m_0^k remain bounded for all k, and the harmonic discrepancy $\delta_r^{\mathrm{harm}} = c_r \cdot m_0^{\lfloor r/2\rfloor-1}$ is a bounded operator on H_q for every r.*

Proof. The curvature element $m_0 = \kappa \cdot \omega_g$ is a scalar operator (multiplication by a cohomology class) tensored with the identity on the state space. As such, its operator norm on H_q is $|\kappa| \cdot \|\omega_g\|$, independent of q. Powers are similarly bounded: $\|m_0^k\| = |\kappa|^k \cdot \|\omega_g\|^k$. The harmonic discrepancy δ_r^{harm} is a finite sum of such terms with coefficients $c_r(\mathcal{A})$ determined by the OPE data; each term is bounded. □

The proposition shows that the curvature does not disrupt the trace-class property of the sewing operators: the HS-sewing condition and the curved differential coexist in the completed Hilbert space. The coderived category provides the correct homological framework; the HS-sewing condition provides the correct functional-analytic framework; together they deliver convergent genus-g partition functions for curved bar complexes.

Proposition 8.17 (Genus-g partition function as coderived trace). *Under HS-sewing, the genus-g partition function is simultaneously:*

(a) *the trace of a trace-class operator on H_q (analytic);*
(b) *an evaluation of the MC element $\Theta_{\mathcal{A}}$ on $[\overline{\mathcal{M}}_g]$ (algebraic);*
(c) *a coderived invariant of the curved bar complex $B^{(g)}(\mathcal{A})$ in D^{co} (homological).*

All three produce the same complex number on the domain of convergence.

Proof. (a) is Corollary 3.3. (b) is the definition of the MC evaluation (the graph-sum expansion). (c) is Proposition 8.11. The equality of the three numbers follows from the compatibility statement of Proposition 8.11(c). □

Conclusion

The paper defines the sewing envelope, proves a trace-class sewing criterion, computes the abelian Heisenberg and lattice models, and separates the remaining analytic questions. The sewing envelope is explicit for the abelian models. For interacting algebras it remains an abstract completion unless a Hilbert-space identification is supplied. Metric independence and the BV/bar comparison require separate estimates: HS-sewing alone does not prove them. The coderived category is the homological target for curved genus-g bar complexes under the stated coacyclicity hypotheses.

Appendix A. Functional analysis background

The functional-analytic facts used above are the following.

A.1. Trace-class and Hilbert–Schmidt operators.

Let H be a separable Hilbert space.

Definition A.1. A bounded operator $T : H \to H$ is:

(a) *Hilbert–Schmidt* if $\|T\|_{\mathrm{HS}}^2 := \sum_j \|Te_j\|^2 < \infty$ for some (hence every) orthonormal basis $\{e_j\}$;
(b) *trace class* if $\|T\|_1 := \sum_j \langle e_j, |T| e_j \rangle < \infty$ where $|T| = (T^*T)^{1/2}$;
(c) *nuclear* if it admits a representation $T = \sum_j \lambda_j \langle f_j, \cdot \rangle g_j$ with $\sum_j |\lambda_j| < \infty$.

For Hilbert spaces, nuclear and trace class agree. In general, trace class $\subset$ Hilbert–Schmidt $\subset$ compact, and the first inclusion is strict even for positive operators: the diagonal operator $\mathrm{diag}(1/n)$ on ℓ^2 is positive and Hilbert–Schmidt but not trace class.

Proposition A.2 (Composition and trace)**.**

(i) *The composition of two Hilbert–Schmidt operators is trace class.*
(ii) *If T is trace class, then $\mathrm{Tr}(T) = \sum_j \langle e_j, Te_j \rangle$ converges absolutely and is independent of the basis.*
(iii) *If T is Hilbert–Schmidt and S is bounded, then ST and TS are Hilbert–Schmidt.*

These standard facts (see Reed–Simon [8]) underpin the trace-class theorem (Proposition 3.2): each pair of pants contributes a Hilbert–Schmidt operator; each sewing circle composes two Hilbert–Schmidt operators into a trace-class operator.

A.2. Fredholm determinants.

Definition A.3. For a trace-class operator K on a separable Hilbert space, the *Fredholm determinant* is

$$\det(1 - K) \;=\; \prod_{j \geq 1} (1 - \lambda_j),$$

where $\{\lambda_j\}$ are the eigenvalues of K counted with algebraic multiplicity. The product converges absolutely whenever K is trace class.

Proposition A.4 (Properties of the Fredholm determinant)**.** *Let K be trace class.*

(i) $|\det(1 - K)| \leq \exp(\|K\|_1)$.
(ii) $\log\det(1 - K) = -\sum_{m \geq 1} \frac{1}{m} \mathrm{Tr}(K^m)$ *whenever* $\|K\| < 1$.
(iii) $\det(1 - K) \neq 0$ *if and only if* $1 \notin \mathrm{spec}(K)$.
(iv) *The map* $K \mapsto \det(1 - K)$ *is continuous in the trace norm.*

For the Heisenberg sewing, the operator K_g has eigenvalues q^n $(n \geq 1)$ at genus 1; the Fredholm determinant is $\det(1 - K_1) = \prod_{n \geq 1} (1 - q^n) = q^{-1/24} \eta(q)$.

A.3. The Bergman space.

Definition A.5. The *Bergman space* $A^2(\Omega)$ of a bounded domain $\Omega \subset \mathbb{C}$ is the Hilbert space of square-integrable holomorphic functions:

$$A^2(\Omega) \;=\; \left\{ f \in \mathcal{O}(\Omega) : \|f\|^2 := \int_\Omega |f|^2 \, dA < \infty \right\}.$$

The Bergman kernel $K_\Omega(z, w) = \sum_j e_j(z) \overline{e_j(w)}$ (with $\{e_j\}$ any orthonormal basis) is the reproducing kernel: $f(z) = \int_\Omega K_\Omega(z, w) f(w) \, dA(w)$.

For the unit disk, $K_D(z,w) = \frac{1}{\pi}(1 - z\bar{w})^{-2}$ and the orthonormal basis is $e_n(z) = \sqrt{(n+1)/\pi}\, z^n$ for $n \geq 0$. The Bergman space of the disk is the one-particle Hilbert space for the Heisenberg sewing (Definition 4.2).

Proposition A.6 (Conformal invariance)**.** *If $\varphi\colon \Omega_1 \to \Omega_2$ is a biholomorphism, then the map $f \mapsto (f \circ \varphi) \cdot \varphi'$ is a unitary isomorphism $A^2(\Omega_2) \xrightarrow{\sim} A^2(\Omega_1)$. In particular, the Bergman space of any simply connected domain is unitarily isomorphic to $A^2(D)$.*

This conformal invariance is the mechanism behind the metric independence of the Heisenberg sewing envelope: any choice of conformal collar gives an isomorphic Bergman space.

A.4. **Siegel modular forms.** The Siegel upper half-space of degree g is

$$\mathfrak{H}_g \;=\; \{\Omega \in \mathrm{Mat}_{g\times g}(\mathbb{C}) : \Omega^T = \Omega,\ \operatorname{Im}\Omega > 0\}.$$

The symplectic group $\mathrm{Sp}_{2g}(\mathbb{Z})$ acts on $\mathfrak{H}_g$ by $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \cdot \Omega = (A\Omega + B)(C\Omega + D)^{-1}$.

Definition A.7. The *Siegel theta function* of an even lattice Λ of rank r is

$$\Theta_\Lambda(\Omega) \;=\; \sum_{\lambda \in \Lambda^g} e^{i\pi\lambda^T\Omega\lambda}, \qquad \Omega \in \mathfrak{H}_g.$$

For unimodular Λ, Θ_Λ is a Siegel modular form of weight $r/2$ with respect to $\mathrm{Sp}_{2g}(\mathbb{Z})$.

At genus 1, $\mathfrak{H}_1 = \mathfrak{H}$ is the classical upper half-plane, $\Omega = \tau$, and the Siegel theta function reduces to the Jacobi theta function $\Theta_\Lambda(\tau) = \sum_{\lambda\in\Lambda} q^{\langle\lambda,\lambda\rangle/2}$ with $q = e^{2\pi i\tau}$.

The convergence of the Siegel theta function on $\mathfrak{H}_g$ (Proposition 5.6) is the lattice analogue of the trace-class property of the Heisenberg sewing operator: both provide absolute convergence of the partition function, by complementary mechanisms.

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